FORMAL RESEARCH EDITION

Matter, Gases and VOTKP

Boundary correlations, gas inverses and the information a record keeps

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3 November 2026 · Prepared for Open Math OS review

$$A - A^2 = (C - C^2)_{AA} + BB^*, \qquad 4\,\mathrm{Tr}(BB^*) \le I_G(A:B).$$

The information bound assumes a gauge-invariant fermionic Gaussian state; the chapter states its complete domain.

ABSTRACT

This edition develops a finite mathematical bridge between boundary correlations, atomic threshold coordinates, specified gas equilibria and retained response. For a positive one-body contraction it proves an exact defect split, sharp order and stability bounds, and a Gaussian mutual-information estimate. It proves finite Saha neutrality and a full inverse threshold reconstruction with temperature-dependent partition corrections. Positive matched-moment and equal-spectrum examples specify what a reduced record can fail to determine. An exact Petermann average is distinguished from a filtered spatial-exit claim; counterexamples and numerical sensitivity are retained. A separate core of 34 Lean theorems compiles, with five false controls rejected. Complete source history, explicit assumptions and a source-integrity audit accompany the work. These are mathematical and declared-model results, not new laboratory measurements or a certification of every interpretation in the historical notes.

Analytic proofs · 34 compiled Lean theorems · Negative controls

Original sources and disputed interpretations kept distinguishable

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1 Source integrity and scope of this edition

During preparation, the author supplied an admission from another assistant that a separate ledger investigation had substituted populations, a single clock and standard bookkeeping for the intended connection-energy framework. The affected run names supplied were E9 repairs, the Heegner helper, midrange runs, split-at-zeros runs, and rally/commutator runs. The quoted duration and usage allocation have not been independently verified here.

Those surrogate tests cannot establish failure of the author's intended framework. This edition does not adopt that inference. The local audit did not identify the named disputed runs, but it did identify an inherited framing problem: the preceding review called standard-model additions a "repair" or "replacement thesis." This edition withdraws any implication that those additions replace or correct the complete Direction–Spin–Throughput/connection-energy framework. The exact source-dependency audit appears later in this document and is preserved in `review/contamination_dependency_audit.md`. Historical notes are retained for provenance, not promoted to authority merely because they are in Drive or carry a passing-run label. No source in Drive or GitHub was changed by this preparation.

The finite matrix, ideal Saha, finite-channel and free-streaming models below are explicitly defined mathematical objects. Their proofs establish statements about those objects. They do not silently redefine the Direction–Spin–Throughput package, tennis-racket/two-books constructions, or the connection-energy note. In particular, the preserved Hermite/BGK counterexample concerns its specified operator and norm, and the closed-Hermitian commutator observation concerns that chosen Hamiltonian; neither identifies the author's connection-energy observable. A claim that one of these models implements that broader framework would require a separate, explicit correspondence checked against the authoritative source. No such correspondence is certified by this edition.

The original matter and VOTKP versions, the exact incoming commit, the new proof sources and their execution records remain separately identifiable. A bibliographic connection is distinguished from a proof dependency. This preserves the material while preventing a disputed historical interpretation from becoming a premise of the new mathematics.

2 Purpose, claims and conventions

This research edition develops the finite mathematical core connecting *Matter at a Scale*, VOTKP, the thermal gas review, and the incoming *Matter on the Smith disk* calculations. It preserves the original sources and failed controls. It adds complete analytic arguments, a separately identified Lean-checked core, independently rerun numerical examples, and explicit counterexamples to stronger readings.

The objects are kept distinct. A positive one-body covariance, an atomic removal-threshold ladder, a gas population distribution, a biorthogonal projector, and a channel-response measure are not interchangeable. They can share coordinate maps or moment identities without becoming the same physical object. The question throughout is precise: **which record determines which later quantity, under which hypotheses?**

2.1 Source lineage and authorship

The programme and original research sources are Jeromie Beasley's. This edition was developed with AI assistance for derivation, code, formalization and adversarial checking. The author should review it before submission or public release. No external peer review or acceptance is implied. Standard matrix identities, Gaussian-state formulas, Saha equilibrium and finite-moment constructions are identified as such; their use together here is not presented as a priority claim.

The source versions are *Matter at a Scale*, v5, DOI 10.5281/zenodo.22019893, and *VOTKP*, v2.9, DOI 10.5281/zenodo.23340757. The preceding review and its full reproduction archive are preserved. The incoming disk source is the exact GitHub commit `881386a0ab00d917af6334a01423fb8b22469f5e` on `claude/matter-disk` in dicipler-pixel/votkp-lean. Twelve related Drive records, their original spreadsheet, both repository figures and all tracked scripts are included in the code archive. The original records are unchanged.

2.2 Evidence classes

Class	Meaning in this edition
Analytic theorem	Complete finite mathematical argument, with its hypotheses stated.
Lean-checked theorem	Exact source declaration compiled against the pinned toolchain; coverage is specified in the formalization inventory.
Model computation	A finite calculation within the stated equations; no new laboratory observation.
Published-data reproduction	Reanalysis of an identified source table with its uncertainty conventions preserved.
Open physical claim	A proposed connection requiring further mathematics, measurement, or an experimentally justified model.

A passing test does not change class by accumulation. The incoming labels `[T]` and `[V]` remain in their historical files, but the present inventory states exactly what was proved and rerun. The formal Lean project is separate from the historical VOTKP library; its success does not certify every old paper or historical script.

2.3 The records being compared

Record	Exact reconstruction or implication	Information it does not supply alone
Positive anchored VOTKP ladder	Every successive removal threshold and binding sum	Gas state, transition weights and preparation
Local covariance block A	Local Gaussian occupations and entropy	How much mixedness is local or crosses the cut
Cross-block matrix B	$X_A = BB^*$ and its boundary-strength spectrum	General higher-order correlations or every later response
Positive finite Saha data	A unique neutral equilibrium in the declared model	Whether a specimen has reached that equilibrium
Specified finite moments	Those moments and derivatives at the origin	A whole characteristic or channel-response function
Retained Hamiltonian, kernel and initial source	Unique retained evolution in the finite block model	A microscopic model from an uncalibrated scalar delay curve

2.4 Conventions

Matrices are finite-dimensional unless a limit is explicitly declared. The order $C \succeq 0$ means positive semidefinite with the standard complex inner product. Matrix star denotes the adjoint. A spatial or retained/hidden decomposition is part of the data. Spectral counts include multiplicity and include equality at the stated threshold. Logs are natural unless a base is displayed. Entropies and mutual information are in nats. A principal complex logarithm has argument in $(-\pi, \pi]$; an arbitrary logarithm branch is not silently substituted.

In the Saha model, N is nucleus number density, n_e is electron number density, and $\beta = 1/(k_B T)$. All energy units must match $k_B T$. In a Hamiltonian toy example with $\hbar = 1$, time is in the reciprocal chosen energy unit. The symbol B in a block matrix is a coupling, while a binding sum is written explicitly or named in its local definition. Collision rate, occupation and frequency are separate parameters even where earlier source files reused a letter.

2.5 Reading route

The boundary chapter establishes the positive-contraction construction and its limits. The disk chapter proves the coordinate relations, including the algebra that survives for non-Hermitian idempotents. The gas chapters establish equilibrium uniqueness, exact input dependence and the inverse peel extraction with partition corrections. The moment and transport chapters give explicit matched-record counterexamples. The Petermann chapter distinguishes an exact band average from an unproved exit criterion. The evidence chapter records the numerical checks, incoming reruns and the new endpoint-filter audit. The final sections give the exact machine-checking coverage and the Open Math OS handoff.

3 A finite-dimensional boundary census for positive contractions

Research setting. This is a theorem-and-proof development prepared for Jeromie Beasley's matter–gas–VOTKP programme on 3 November 2026. The block identities, spectral estimates and Gaussian identities below are standard finite-dimensional mathematics. Their use as a proposed continuation of the paper's pure-projector census is the application developed here; no priority claim is made. A theorem marked **proved here** has a complete mathematical proof below. This label does not mean that the statement has been checked in Lean; the separate formalization inventory is authoritative for that status.

3.1 Definitions and conventions

Let $\mathcal{H} = \mathcal{H}_A \oplus \mathcal{H}_B$ be a fixed finite-dimensional orthogonal decomposition of a complex Hilbert space, with dimensions a, b. Let $C = C^*$ satisfy $0 \le C \le I$. Write

$$C = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \qquad T_A = (C - C^2)_{AA}, \qquad F_A = A - A^2, \qquad X_A = BB^*.$$

Here $B : \mathcal{H}_B \to \mathcal{H}_A$; the label B on this operator is distinct from the subsystem name. Define $X_B = B^*B$. The norm $\|\cdot\|$ is the operator norm; $\|\cdot\|_F$ is the Hilbert–Schmidt norm. Every eigenvalue count below includes algebraic multiplicity (equal to geometric multiplicity for the Hermitian matrices used). At a threshold s, equality is included:

$$\mathcal{N}_Y(s) = \operatorname{rank} \mathbf{1}_{[s,\infty)}(Y).$$

Fix $0 < \varepsilon < 1/2$, put $\theta = \varepsilon(1-\varepsilon) > 0$, and define

$$N_\varepsilon^{\mathrm{raw}}(C; A) = \operatorname{rank} \mathbf{1}_{[\varepsilon, 1-\varepsilon]}(A), \quad N_\varepsilon^{\partial}(C; A) = \mathcal{N}_{X_A}(\theta).$$

The first count concerns intermediate local occupations; the second concerns squared cross-block singular values. The partition is part of the definition. There is no partition-independent phase or substance label here.

3.2 B1. Exact defect decomposition — proved here

For every block matrix above,

$$\boxed{F_A = T_A + X_A.}$$

Proof. The AA block of C^2 is $A^2 + BB^*$. Hence $T_A = A - A^2 - BB^*$. Rearranging gives the identity. Hermiticity and positivity are not needed for this algebraic calculation if the lower-left block is treated as a general independent block. They become essential for the order statements below. □

3.3 B2. Positivity, sharp bounds and singular values — proved here

$$0 \le X_A \le F_A \le \tfrac{1}{4} I_A, \qquad 0 \le T_A \le F_A, \qquad \|B\| \le \tfrac{1}{2}.$$

Proof. Compression preserves order, so $0 \le A \le I_A$. The finite-dimensional spectral theorem applied to C gives $C - C^2 \ge 0$, because its eigenvalues are $c(1-c) \ge 0$ for $c \in [0,1]$. Thus $T_A \ge 0$. Also $X_A = BB^* \ge 0$. B1 yields both $X_A \le F_A$ and $T_A \le F_A$. Finally,

$$F_A = \tfrac{1}{4} I_A - (A - \tfrac{1}{2} I_A)^2 \le \tfrac{1}{4} I_A.$$

Therefore $\|B\|^2 = \|BB^*\| \le 1/4$. Equality is attained for the two-mode rank-one projector $C = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. □

In particular, the eigenvalues of X_A are the squared singular values of B, with zero padding when $a > b$. The threshold can equivalently be written $s_j(B) \ge \sqrt{\varepsilon(1-\varepsilon)}$. Using the unsquared threshold for these singular values would be a different diagnostic.

3.4 B3. Two-sided symmetry and finite-size limits — proved here

The positive eigenvalues of $X_A = BB^*$ and $X_B = B^*B$ agree, including multiplicity. Consequently,

$$N_\varepsilon^\partial(C; A) = N_\varepsilon^\partial(C; B), \qquad N_\varepsilon^\partial \le \operatorname{rank} B \le \min(a, b),$$

$$\operatorname{Tr} X_A = \operatorname{Tr} X_B = \|B\|_F^2 \le \tfrac{1}{4} \min(a, b).$$

Proof. If $BB^*u = \lambda u$ with $\lambda > 0$, then $B^*u \neq 0$ and $B^*B(B^*u) = \lambda B^*u$. The reverse map is B/λ, so the eigenspaces have equal dimensions. The remaining statements follow by counting and summing the positive eigenvalues and using B2. □

The finite census cannot diverge at fixed finite a, b. A divergent per-boundary or scaling-limit census must specify the sequence of spaces and limits.

3.5 B4. Pure-projector recovery and a universal one-sided comparison — proved here

For every positive contraction,

$$N_\varepsilon^\partial(C; A) \le N_\varepsilon^{\mathrm{raw}}(C; A).$$

If $C^2 = C$, equality holds exactly.

Proof. For $\nu \in [0, 1]$,

$$\nu(1-\nu) - \varepsilon(1-\varepsilon) = (\nu - \varepsilon)(1 - \varepsilon - \nu),$$

so $\nu(1-\nu) \ge \theta$ if and only if $\nu \in [\varepsilon, 1-\varepsilon]$. The spectral theorem therefore gives $\mathcal{N}_{F_A}(\theta) = N_\varepsilon^{\mathrm{raw}}$. By B2, $X_A \le F_A$; the min–max principle gives $\lambda_j(X_A) \le \lambda_j(F_A)$ for each ordered eigenvalue. This proves the inequality. If $C^2 = C$, B1 gives $T_A = 0$, hence $X_A = F_A$, proving equality. No numerical threshold tolerance is used in this exact statement. □

3.6 B5. Trace estimate and its exact meaning — proved here

$$\boxed{\theta N_\varepsilon^\partial(C; A) \le \operatorname{Tr} X_A = \sum_{i \in A, j \in B} |C_{ij}|^2.}$$

Proof. Each of the counted nonnegative eigenvalues contributes at least θ to the trace. The uncounted eigenvalues are nonnegative. The entry formula follows from $\operatorname{Tr}(BB^*) = \sum_{ij} |B_{ij}|^2$. □

This is an upper bound on the count, not an identity determining it from the trace. A count of zero allows nonzero eigenvalues below θ; it is not a zero-correlation certificate. By contrast, $\operatorname{Tr} X_A = 0$ if and only if $B = 0$.

3.7 B6. Local-unitary covariance — proved here

Let $U = U_A \oplus U_B$ be block diagonal and unitary, and set $\widetilde{C} = UCU^*$. Then

$$\widetilde{A} = U_A A U_A^*, \quad \widetilde{B} = U_A B U_B^*, \quad \widetilde{X}_A = U_A X_A U_A^*, \quad \widetilde{T}_A = U_A T_A U_A^*.$$

Both censuses and all the displayed spectra and traces are unchanged.

Proof. Block multiplication gives the formulas; unitary conjugation preserves eigenvalues and trace. □

A unitary that mixes the two regions changes the partition-relative observable and is not covered by this invariance.

3.8 B7. Operator-norm stability and a certified threshold margin — proved here

Let $C, \widetilde{C}$ be positive contractions on the same partition and put $\delta = \|C - \widetilde{C}\|$. Then

$$\|X_A(C) - X_A(\widetilde{C})\| \le \delta.$$

For $\delta < \theta$,

$$\mathcal{N}_{X_A(C)}(\theta + \delta) \le N_\varepsilon^\partial(\widetilde{C}; A) \le \mathcal{N}_{X_A(C)}(\theta - \delta).$$

In particular, if

$$\operatorname{dist}(\theta, \operatorname{spec} X_A(C)) > \delta,$$

then $N_\varepsilon^\partial(C; A) = N_\varepsilon^\partial(\widetilde{C}; A)$.

Proof. Write the cross blocks as $B, \widetilde{B}$. Compression gives $\|B - \widetilde{B}\| \le \delta$, while B2 gives $\|B\|, \|\widetilde{B}\| \le 1/2$. Therefore

$$BB^* - \widetilde{B}\widetilde{B}^* = (B - \widetilde{B})B^* + \widetilde{B}(B^* - \widetilde{B}^*)$$

has norm at most $(\|B\| + \|\widetilde{B}\|)\|B - \widetilde{B}\| \le \delta$. The Hermitian min–max principle bounds each eigenvalue change by δ. The threshold inequalities follow by applying those bounds to ordered eigenvalues. The strict distance condition prevents any eigenvalue from reaching or crossing the threshold. □

The assumption $\delta < \theta$ merely keeps the lower comparison threshold positive; the spectral inequalities themselves remain meaningful without it. For floating-point computations, an empirically small residual is not by itself a rigorous operator-norm error bound. A certified numerical count needs a validated error bound and a margin larger than it.

3.9 B8. A near-projector comparison controlled by global mixing — proved here

Put $\tau = \|C - C^2\|$. Then $0 \le \tau \le 1/4$ and

$$X_A \le F_A \le X_A + \tau I_A.$$

Consequently,

$$\boxed{\mathcal{N}_{F_A}(\theta + \tau) \le N_\varepsilon^\partial(C; A) \le \mathcal{N}_{F_A}(\theta) = N_\varepsilon^{\mathrm{raw}}(C; A).}$$

If $\theta + \tau \le 1/4$, define

$$\varepsilon_\tau = \frac{1 - \sqrt{1 - 4(\theta + \tau)}}{2}.$$

Then the lower bound is $N_{\varepsilon_\tau}^{\mathrm{raw}}$, with the endpoint $\varepsilon_\tau = 1/2$ interpreted as the multiplicity of $1/2$. If $\theta + \tau > 1/4$, the lower comparison count is zero.

Proof. Positivity of $C - C^2$ implies $0 \le C - C^2 \le \tau I$. Compressing gives $0 \le T_A \le \tau I_A$. Apply B1 and ordered eigenvalue monotonicity. Solving $\nu(1 - \nu) \ge \theta + \tau$ gives the last assertion. □

This quantifies the gap between local occupation counting and cross-boundary counting without assuming that their matrices commute. It recovers B4 when $\tau = 0$.

3.10 B9. Same local covariance, different boundary correlation — proved here

Every positive contraction A has the projector extension

$$P_A = \begin{pmatrix} A & \sqrt{A - A^2} \\ \sqrt{A - A^2} & I - A \end{pmatrix} \quad \text{on } \mathcal{H}_A \oplus \mathcal{H}_A.$$

It has the same AA block as the disconnected positive contraction $C_0 = A \oplus (I - A)$. However,

$$X_A(P_A) = A - A^2, \qquad X_A(C_0) = 0.$$

Proof. The square root exists by the spectral theorem and commutes with A. The diagonal blocks of P_A^2 are $A^2 + (A - A^2) = A$ and $(A - A^2) + (I - A)^2 = I - A$. The off-diagonal block is $A\sqrt{A - A^2} + \sqrt{A - A^2}(I - A) = \sqrt{A - A^2}$. Thus $P_A^2 = P_A = P_A^*$, so it is a positive contraction. The cross blocks give the asserted X's. □

The local occupation spectrum alone therefore does not determine which part of its defect is boundary correlation and which part is global mixedness. This is a precise limitation of the local record.

3.11 B10. Gaussian number-covariance interpretation — proved here under stated physics assumptions

Let $a_i, a_i^\dagger$ satisfy the finite fermionic canonical anticommutation relations. Suppose ρ is a gauge-invariant quasi-free (number-conserving Gaussian) state. Set

$$C_{ij} = \langle a_j^\dagger a_i \rangle_\rho, \quad n_i = a_i^\dagger a_i, \quad N_A = \sum_{i \in A} n_i, \quad N_B = \sum_{j \in B} n_j.$$

This convention makes one-body evolution under $H = \sum_{ij} h_{ij} a_i^\dagger a_j$ satisfy $C(t) = e^{-iht} C(0) e^{iht}$, with $\hbar = 1$. Then

$$\mathrm{Var}(N_A) = \mathrm{Tr}(A - A^2), \qquad \mathrm{Cov}(N_A, N_B) = -\mathrm{Tr}\, X_A,$$

and hence

$$\mathrm{Var}(N_A) = \mathrm{Tr}\, T_A - \mathrm{Cov}(N_A, N_B).$$

Proof. For $i \neq j$, Wick factorization and the absence of anomalous pair correlations give

$$\langle n_i n_j \rangle = C_{ii} C_{jj} - C_{ij} C_{ji}.$$

For $i = j$, the CAR give $n_i^2 = n_i$. Thus

$$\mathrm{Cov}(n_i, n_j) = \delta_{ij} C_{ii} - |C_{ij}|^2.$$

Sum over $i, j \in A$ for the variance and over $i \in A, j \in B$ for the cross covariance. Apply B1 for the last formula. □

When C is a projector, the state is a pure Slater state, and $\mathrm{Var}(N_A) = \mathrm{Tr}\, X_A$. For a Gibbs Gaussian state, T_A is the global thermal occupation defect restricted to A. For a general mixed Gaussian state it is occupation mixedness; calling it thermal requires a thermodynamic preparation. The matrix $A - A^2$ itself is not a scalar variance; its trace is.

3.12 B11. Counterexamples outside the claimed scope — proved here

Disconnected thermal state. If $C = pI$ with $\varepsilon \leq p \leq 1 - \varepsilon$, then $N_\varepsilon^{\mathrm{raw}} = a$, while $X_A = 0$ and $N_\varepsilon^\partial = 0$. The associated Gaussian state is a product over modes, so its mutual information across the partition is zero. Its local number variance is $ap(1-p)$.

Non-Gaussian classical correlations invisible to C. On two modes, the state

$$\rho_{\mathrm{cl}} = \tfrac{1}{2}|10\rangle\langle 10| + \tfrac{1}{2}|01\rangle\langle 01|$$

has $C = \frac{1}{2} I_2$, hence $X_A = 0$, but $\mathrm{Cov}(n_1, n_2) = -1/4$ and mutual information $\log 2$. The Gaussian state $I_4/4$ has the same C, but zero covariance and zero mutual information. This directly disproves B10 without its quasi-free assumption.

Entanglement invisible to C. On four modes split as $A = \{1, 2\}$, $B = \{3, 4\}$, let

$$|\Psi\rangle = \frac{|1100\rangle + |0011\rangle}{\sqrt{2}}.$$

It has fixed total particle number two and $C = \frac{1}{2} I_4$, so $X_A = 0$. Its reduced state on A has eigenvalues $1/2, 1/2$, hence entanglement entropy $\log 2$. Each branch has even local parity, so the example does not rely on a local even/odd superposition. The one-body off-diagonal elements vanish because no one-body operator connects the two occupation configurations. Thus the boundary census is not a universal entanglement detector.

3.13 B12. Even the full boundary-strength record does not determine later response — proved here

Consider two modes and

$$C_{\pm} = \tfrac{1}{2} I_2 \pm r\sigma_y, \quad 0 < r \leq \tfrac{1}{2}, \quad h = \omega\sigma_x, \quad \omega > 0.$$

Both matrices are positive contractions. Their local blocks are $A = D = 1/2$, their global eigenvalues are $1/2 \pm r$, and both have $X_A = X_B = r^2$. Thus both raw and boundary censuses, the full cross-block singular-value list, all occupation spectra, and their Gaussian mutual information agree initially. Under the same number-conserving one-body Hamiltonian,

$$C_{\pm}(t) = \tfrac{1}{2} I_2 \pm r\big(\cos(2\omega t)\sigma_y + \sin(2\omega t)\sigma_z\big),$$

so

$$\boxed{\langle n_A \rangle_{\pm}(t) = \tfrac{1}{2} \pm r \sin(2\omega t).}$$

Proof. The Pauli matrix σ_y has eigenvalues ± 1, proving positivity and the spectra. Its off-diagonal elements are $-i, i$, so multiplying each cross block by its adjoint gives r^2. Conjugation by $e^{-i\omega t\sigma_x}$ gives the stated rotation; alternatively differentiate and use $[\sigma_x, \sigma_y] = 2i\sigma_z$. The AA entry gives the occupation. □

The missing datum is a relative phase with respect to the fixed coupling Hamiltonian. The initial currents are **not** matched: the displayed occupations have initial derivatives $\pm 2\omega r$. Thus this is a matched occupation/spectral/boundary-strength example, not the matched density/flow/stress example supplied separately by the kinetic construction. Although a local unitary maps the two initial covariances into each other, using that transformation as a mere change of coordinates would also transform the Hamiltonian. Here the Hamiltonian is held fixed, so the two preparations have physically different phase relative to it. This is an exact observability limitation, not a claim that a census should predict all dynamics.

3.14 B13. Quantitative Gaussian mutual-information bounds — proved here

Define the binary entropy in natural logarithms by

$$h(x) = -x \log x - (1-x)\log(1-x), \qquad h(0) = h(1) = 0.$$

For the gauge-invariant quasi-free state of B10, its bipartite mutual information is

$$I_G(A:B) = \operatorname{Tr} h(A) + \operatorname{Tr} h(D) - \operatorname{Tr} h(C).$$

Then

$$\boxed{4 \operatorname{Tr} X_A \leq I_G(A:B).}$$

If, in addition, $0 < \delta \leq 1/2$ and

$$\delta I \leq C \leq (1-\delta) I,$$

then

$$\boxed{I_G(A:B) \le \frac{\operatorname{Tr} X_A}{\delta(1-\delta)}.}$$

Proof of the entropy formula. Diagonalizing the covariance of a gauge-invariant quasi-free state gives independent fermionic occupation modes with probabilities equal to its eigenvalues. Their Bernoulli entropies add, giving $S(\rho) = \operatorname{Tr} h(C)$. Reduced states remain quasi-free with the compressed covariances A, D, giving the displayed mutual-information formula. Values zero and one follow by continuity, or by including deterministic occupation modes.

Trace Jensen fact. If a scalar function f is concave on an interval and the spectra of Hermitian matrices H, K lie in it, then

$$\operatorname{Tr} f(tH + (1-t)K) \ge t \operatorname{Tr} f(H) + (1-t) \operatorname{Tr} f(K) \quad (0 \le t \le 1).$$

To see this directly, diagonalize $tH+(1-t)K$ with an orthonormal basis u_i. Apply scalar concavity to the two numbers $\langle u_i, Hu_i\rangle$ and $\langle u_i, Ku_i\rangle$. Then scalar Jensen applied to the spectral weights of H gives $f(\langle u_i, Hu_i\rangle) \ge \langle u_i, f(H)u_i\rangle$, and likewise for K. Sum over i. For convex f, every inequality reverses. No operator-concavity assumption is being made.

Proof of the lower bound. Let

$$S = I_A \oplus (-I_B), \qquad C_0 = A \oplus D = \tfrac{1}{2}(C + SCS).$$

The scalar function $f(x) = h(x) + 2x^2$ is concave on $[0,1]$, because in its interior

$$f''(x) = 4 - \frac{1}{x(1-x)} \le 0,$$

and it extends continuously to the endpoints. Trace Jensen and the unitary invariance of trace give $\operatorname{Tr} f(C_0) \ge \operatorname{Tr} f(C)$. Consequently,

$$I_G(A:B) \ge 2[\operatorname{Tr} C^2 - \operatorname{Tr} C_0^2] = 4 \operatorname{Tr}(BB^*).$$

Proof of the upper bound. Compression gives the same $[\delta, 1-\delta]$ spectral bounds for C_0. On this interval,

$$g(x) = h(x) + \frac{x^2}{2\delta(1-\delta)}$$

is convex, since

$$g''(x) = \frac{1}{\delta(1-\delta)} - \frac{1}{x(1-x)} \ge 0.$$

Reversed trace Jensen yields $\operatorname{Tr} g(C_0) \le \operatorname{Tr} g(C)$. Rearranging and using $\operatorname{Tr} C^2 - \operatorname{Tr} C_0^2 = 2 \operatorname{Tr} X_A$ gives the bound. At $\delta = 1/2$, the only allowed covariance is $C = I/2$, and both sides vanish. □

Sharpness and consequences. The lower coefficient 4 is optimal: for $C = \frac{1}{2}I_2 + r\sigma_x$, with $r \to 0$, one has $\operatorname{Tr} X_A = r^2$ and

$$I_G = 2\log 2 - 2h(\tfrac{1}{2} + r) = 4r^2 + O(r^4).$$

Combining B5 and B13 gives the rigorous census bound

$$\boxed{N_\varepsilon^\partial(C;A) \le \frac{I_G(A:B)}{4\varepsilon(1-\varepsilon)}.}$$

Within this Gaussian class, $I_G = 0$ if and only if $X_A = 0$: the lower bound gives one direction, and the block-diagonal covariance and entropy formula give the other. This equivalence concerns mutual information, not entanglement. The finite-temperature upper estimate additionally requires a uniform occupation margin δ; it is not a uniform bound near a pure state.

An independently established Gaussian mutual-information area bound would therefore imply a census area bound with the displayed constant. This conditional implication does not transfer a bosonic theorem to fermions, or establish an area bound by itself. The non-Gaussian counterexamples B11 remain decisive outside the stated class. For entropy measured in bits, divide the natural-log information bounds by $\log 2$.

3.15 Dependency and formalization guide

Result	Inputs	Proof mechanism	Scope
B1	Block multiplication	Ring/matrix algebra	No positivity needed
B2	Hermitian positive contraction	Spectral theorem; square completion	Finite-dimensional
B3	Rectangular cross block	Positive-eigenvalue eigenspace correspondence	Finite-dimensional
B4	B1–B2; scalar factorization	Min–max and functional calculus	Exact threshold convention
B5	Nonnegative eigenvalue list	Sum bound	Does not identify MI
B6	Partition-preserving unitary	Matrix conjugation	Partition held fixed
B7	B2; compression norm bound	Product difference; eigenvalue stability	Certified margin required
B8	B1; defect norm	Order and min–max	No commutativity assumption
B9	Positive contraction A	Positive square root	Shows missing global information
B10	CAR; gauge-invariant quasi-free state	Wick factorization	Excludes unmodeled pairing/non-Gaussianity
B11	Explicit density operators	Direct evaluation	Counterexamples to overextension
B12	Explicit Pauli matrices	Exact unitary conjugation	Dynamical phase is extra information
B13	Gauge-invariant quasi-free state; scalar entropy curvature	Trace Jensen and block pinching	Mutual-information bounds, not general entanglement claims

No result here identifies an ionization threshold with a many-body gap, derives a cosmological mechanism, or proves the original paper's thermodynamic asymptotics. The proposed census can be studied alongside those questions without replacing their missing hypotheses.

4 What the boundary coordinate on the Smith disk proves

This note integrates the actual incoming `matter-disk` work at commit `881386a0ab00d917af6334a01423fb8b22469f5e`, especially README §7 and `scripts/s07_boundary_census_disk.py`. It preserves the algebraic extension to biorthogonal idempotents while distinguishing it from the positive-covariance interpretation. The new section is an addition to the source record; the incoming files are retained unchanged.

4.1 D1. The positive-contraction boundary coordinate — proved here

Under B1–B2, each eigenvalue λ of $X_A = C_{AB}C_{BA} = BB^*$ lies in $[0, 1/4]$. Therefore

$$\Gamma_b(\lambda) = \sqrt{1 - 4\lambda} \in [0, 1]$$

is a real, unsigned coordinate, and

$$\lambda = \frac{1 - \Gamma_b^2}{4}.$$

For $0 < \varepsilon < 1/2$,

$$\boxed{\lambda \geq \varepsilon(1-\varepsilon) \iff \Gamma_b(\lambda) \leq 1 - 2\varepsilon.}$$

Proof. The square root is real and nonnegative by B2. Both sides of $\Gamma_b \leq 1 - 2\varepsilon$ are nonnegative. Squaring and rearranging proves the equivalence. □

For a pure **orthogonal** projector, B4 identifies $\lambda = \nu(1-\nu)$, hence

$$\Gamma_b = \sqrt{1 - 4\nu(1-\nu)} = |1 - 2\nu|.$$

The signed occupation coordinate $\Gamma = 1 - 2\nu$ is recovered only up to sign. This is equality with its modulus, not equality with the original signed disk point. The map forgets the distinction between ν and $1 - \nu$.

For mixed positive contractions, X_A generally does not commute with A. Its eigenvalues cannot in general be paired level by level with $\nu(1-\nu)$; the order and count comparisons B4/B8 remain valid. The coordinate lies on a real half-diameter of the Smith disk. Its definition alone provides neither a complex phase nor the disk's full geometry.

4.2 D2. What "back to the rim" means — proved here

If $C = pI$, then $X_A = 0$, so every $\Gamma_b = 1$, although the raw occupation coordinate is $1 - 2p$. At $p = 1/2$, all raw coordinates are zero while all boundary coordinates are one. This is the disconnected product-state control.

More generally, a zero boundary count only proves $\Gamma_b > 1 - 2\varepsilon$ for each level; it does **not** prove $\Gamma_b = 1$ or $X_A = 0$. The original 128-site uniform-ring result at $T = 0.5$ has zero boundary count at $\varepsilon = 0.1$ but $\operatorname{Tr} X_A \approx 0.170835 > 0$. Its boundary correlations are below that counting threshold, not absent. The warm-state wording must distinguish the exact disconnected control from an interacting or hopping system with small surviving cross correlations.

4.3 D3. The idempotent algebra survives opening — proved here

Now drop Hermiticity and positivity. On a finite complex vector space with a fixed block decomposition, let

$$C = \begin{pmatrix} A & B \\ E & D \end{pmatrix}, \qquad C^2 = C.$$

Define the algebraic cross product $Y_A = BE = C_{AB}C_{BA}$, which need not equal BB^*. Then

$$\boxed{Y_A = A - A^2.}$$

If $\nu_1, \dots, \nu_a$ are the eigenvalues of A, with algebraic multiplicity, the eigenvalues of Y_A are

$$\lambda_j = \nu_j(1 - \nu_j), \qquad j = 1, \dots, a,$$

again with algebraic multiplicity. This statement allows nontrivial Jordan blocks.

Proof. The AA block of idempotence gives $A^2 + BE = A$. Triangularize A over $\mathbb{C}$. A polynomial in that triangular matrix has diagonal entries given by the polynomial evaluated on its diagonal. Taking the polynomial $p(z) = z - z^2$ proves the spectral multiset statement, without assuming diagonalizability of A. □

This is the precise algebraic content that the incoming §7 opened-chain test verifies numerically. It legitimately carries over to a biorthogonal projector. Positivity bounds and Gaussian-state interpretations do not follow from this algebra alone.

4.4 D4. A branch-independent radial census survives too — proved here

For a complex eigenvalue λ, define

$$R_b(\lambda) = \sqrt{|1 - 4\lambda|} \in [0, \infty).$$

For every idempotent in D3,

$$\boxed{R_b(\nu(1-\nu)) = |1 - 2\nu|.}$$

Consequently, for any $r \geq 0$, the two multisets give the identical radial count

$$\#\{j : R_b(\lambda_j) \leq r\} = \#\{j : |1 - 2\nu_j| \leq r\}.$$

Proof. The identity $1 - 4\nu(1-\nu) = (1-2\nu)^2$ gives

$$\sqrt{|1 - 4\nu(1-\nu)|} = \sqrt{|1-2\nu|^2} = |1 - 2\nu|.$$

Apply D3's spectral multiset mapping. □

The radial condition is equivalently

$$\left|\lambda - \tfrac{1}{4}\right| \leq \tfrac{r^2}{4},$$

a disk in the complex λ-plane. In the positive-contraction case, restricting λ to $[0, 1/4]$ reduces this to the one-sided real inequality used in D1. For general non-Hermitian idempotents, there is no ordered comparison $\lambda \geq \varepsilon(1-\varepsilon)$ for complex λ.

Thus the incoming statement can be repaired precisely: **the idempotent spectral relation and the radial disk count carry over unchanged after using R_b; the PSD census, variance interpretation and real-square-root restriction do not.** Taking a principal complex square root of $(1-2\nu)^2$ gives one of $\pm(1-2\nu)$, not its modulus. Taking its absolute value, or using the branch-independent formula above, gives the radial coordinate.

4.5 D5. Idempotence alone cannot supply positivity — proved here

The real matrix

$$C = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}$$

satisfies $C^2 = C$. Its compression is $A = 2$, and its algebraic cross product is $Y_A = -2$, which is negative. The radial coordinate is $R_b = 3$, outside the unit disk. If one instead substitutes $BB^* = 1$, the projector identity $Y_A = A - A^2$ is lost. These two cross products cannot be exchanged silently.

More generally, **every** square complex matrix A is the compression of the idempotent

$$\mathcal{P}(A) = \begin{pmatrix} A & I \\ A - A^2 & I - A \end{pmatrix}.$$

Proof. Multiply blocks. The upper-left block of $\mathcal{P}(A)^2$ is $A^2 + A - A^2 = A$; the upper-right is $A + I - A = I$; the lower-left is $(A - A^2)A + (I - A)(A - A^2) = A - A^2$, since all the terms are polynomials in A; the lower-right is $A - A^2 + (I - A)^2 = I - A$. □

Hence global idempotence places no positivity or spectral-interval restriction on a general compression. A positive-metric reformulation would require its own declared inner product, adjoint, orthogonal partition and physical interpretation. It is not obtained merely by relabeling a biorthogonal projector as a density covariance.

4.6 D6. Cayley-domain statement with the logarithm branch made explicit — proved here

For $z \in \mathbb{C} \setminus \{-1\}$, set $\Gamma = (z-1)/(z+1)$. Then

$$|z+1|^2 - |z-1|^2 = 4\,\mathrm{Re}\, z,$$

so

$$|\Gamma| < 1 \iff \mathrm{Re}\, z > 0, \quad |\Gamma| = 1 \iff \mathrm{Re}\, z = 0, \quad |\Gamma| > 1 \iff \mathrm{Re}\, z < 0.$$

For $z = e^W \neq -1$, the Cayley expression is $\tanh(W/2)$, and

$$|\tanh(W/2)| < 1 \iff \cos(\mathrm{Im}\, W) > 0.$$

If $W = \mathrm{Log}\, z$ is the principal logarithm with imaginary part in $(-\pi, \pi]$, this is equivalent to

$$|\,\mathrm{Im}\, W| < \pi/2.$$

Proof. Expand the squared moduli. Divide their difference by the positive $|z+1|^2$, then use $\operatorname{Re} e^W = e^{\operatorname{Re} W}\cos(\operatorname{Im} W)$. On the stated principal branch, cosine is positive exactly in the indicated open interval. □

Without the branch restriction, the interval statement is false: the criterion is periodic modulo 2π. At $z=-1$ the Cayley coordinate has a pole and is undefined, rather than an ordinary point outside the disk. For a real occupation $0<\nu<1$, the odds $z=(1-\nu)/\nu$ give $\Gamma = 1-2\nu$; the endpoints are obtained by continuous limits.

4.7 D7. Coordinate stability and the nonnormal boundary of the estimate — proved here

For $\lambda, \tilde{\lambda} \in [0, 1/4]$,

$$|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| \le 2\sqrt{|\lambda - \tilde{\lambda}|}.$$

If both radii are at least $g > 0$, the stronger bound is

$$|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| \le \frac{2}{g}|\lambda - \tilde{\lambda}|.$$

Proof. For nonnegative x, y, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$. Apply this with $x = 1-4\lambda$, $y = 1-4\tilde{\lambda}$. Alternatively, rationalizing gives the exact identity

$$|\Gamma_b(\lambda) - \Gamma_b(\tilde{\lambda})| = \frac{4|\lambda - \tilde{\lambda}|}{\Gamma_b(\lambda) + \Gamma_b(\tilde{\lambda})},$$

when the denominator is nonzero, proving the second bound. If both radii vanish the first bound is immediate. □

Combining D7 with B7 gives a $2\sqrt{\delta}$ coordinate bound for corresponding ordered Hermitian boundary eigenvalues when $\|C - \widetilde{C}\| \le \delta$. The square-root loss near the center is real: changing $\lambda = 1/4$ to $1/4 - h$ changes its radius from zero to $2\sqrt{h}$.

The scalar radial map R_b obeys the same one-half Hölder estimate for complex λ, by the reverse triangle inequality. However, B7's eigenvalue perturbation step is a Hermitian result and cannot be transferred to nonnormal matrices. For example,

$$J_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad J_\delta = \begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}$$

differ in norm by δ, but their eigenvalues change from $0, 0$ to $\pm\sqrt{\delta}$. By D5, these matrices can themselves be embedded as compressions of idempotents. A computed non-Hermitian residual therefore needs a conditioning or pseudospectral analysis before it becomes a certified disk count. Idempotence alone does not supply that analysis.

4.8 What this joins and what it does not identify

The occupation odds, boundary-strength radius and gas-population odds share exact coordinate identities. Their underlying records remain different: an eigenvalue of a local covariance, a cross-boundary singular strength, and a charge-population ratio are distinct observables. These maps create no missing matrix elements, history labels or experimental data. Their value is a precise common language in which the hypotheses and lost information can be written and tested.

5 Atomic records, finite Saha closure, and resolved response

Formal mathematical supplement for Jeromie Beasley's Matter–Gases–VOTKP project, 3 November 2026. This document proves finite-dimensional and finite-charge statements under explicit hypotheses. It does not identify ionization thresholds with collective gaps, establish local thermodynamic equilibrium in a specimen, or infer a gas memory kernel from the xenon delay table. These are standard mathematical constructions developed into a precise account of the proposed bridge; no priority claim is made.

The original source records are Beasley, *Matter at a Scale*, v5, DOI 10.5281/zenodo.22019893, and *VOTKP*, v2.9, DOI 10.5281/zenodo.23340757. The companion deep-dive archive preserves the original sources, the helium input rows with their grades, and the retrospective xenon source-table reproduction.

5.1 1. Positive threshold records

Fix an integer $Z \geq 1$. A complete positive threshold ladder is $I = (I_0, \dots, I_{Z-1}) \in (0,\infty)^Z$. The index labels successive relaxed charge-state removals. It is not an excitation-energy label within one fixed-charge Hamiltonian. Define, for $0 \leq q < Z-1$,

$$\Gamma_q = \frac{I_{q+1} - I_q}{I_{q+1} + I_q}.$$

Theorem G1 — anchored coordinate equivalence. The map

$$I \longmapsto (I_0, \Gamma_0, \dots, \Gamma_{Z-2})$$

is a bijection from $(0,\infty)^Z$ onto $(0,\infty) \times (-1,1)^{Z-1}$. Its inverse is

$$I_q = I_0 \prod_{j=0}^{q-1} \frac{1+\Gamma_j}{1-\Gamma_j}.$$

Proof. Positivity gives $-1 < (b-a)/(b+a) < 1$ for every $a, b > 0$. Direct algebra gives $(1+\Gamma)/(1-\Gamma) = b/a$. Multiplying adjacent ratios telescopes. Conversely, for any positive anchor and ratios in $(-1,1)$, every factor in the displayed inverse is positive, and the adjacent Cayley transform recovers the supplied ratio. For $Z = 1$, the ratio vector is empty and the positive anchor is the whole record. □

Corollary G1a — scale quotient and binding sum. Positive common scaling $I \mapsto cI$ leaves every Γ_q unchanged. Two connected positive ladders have identical ratio vectors if and only if they differ by one positive common scale. The anchored vector determines the remaining electronic binding sums $B_q = \sum_{j=q}^{Z-1} I_j$.

Proof. Common scaling cancels in each ratio. Equal ratios give equal adjacent quotients, hence by induction $I'_q/I_q = I'_0/I_0$. The binding sums are functions of the reconstructed ladder. □

Information consequence. Any function of the complete threshold ladder can be re-expressed as a function of the complete anchored vector, and conversely. A change of coordinates can change numerical conditioning or a chosen statistical representation; it cannot add physical information. Neither representation determines a specimen's temperature, density, partition weights, transition matrix elements, collision rates or preparation. Missing ladder entries remain missing data.

5.2 2. Finite Saha closure

Fix $Z \geq 1$, a nucleus number density $N > 0$, and positive coefficients $S_0, \dots, S_{Z-1} > 0$. These coefficients have number-density units. The abstract finite Saha model is

$$\frac{n_{q+1}}{n_q} = \frac{S_q}{x}, \qquad \sum_{q=0}^{Z} n_q = N, \qquad x = \sum_{q=0}^{Z} q n_q,$$

where $x = n_e > 0$ is the electron number density. All charge states from 0 to Z are included. The final condition is charge neutrality for a pure species with no imposed background charge. Mixtures, externally supplied electrons and non-LTE kinetics require different closure equations.

Set $a_0 = 1$, $a_q = \prod_{j=0}^{q-1} S_j$, and for $x > 0$ define

$$W(x) = \sum_{q=0}^{Z} a_q x^{-q}, \qquad p_q(x) = \frac{a_q x^{-q}}{W(x)}, \qquad \mu(x) = \sum_q q p_q(x), \qquad V(x) = \sum_q (q - \mu(x))^2 p_q(x).$$

Every population is strictly positive. Therefore $0 < \mu(x) < Z$ and $V(x) > 0$. The population ratios and normalization hold by construction, so neutrality is equivalent to $x = N\mu(x)$.

Lemma G2 — exponential-family derivative. Writing $y = \log x$,

$$\frac{dp_q}{dy} = p_q(\mu - q), \qquad \frac{d\mu}{dy} = -V.$$

More generally, for a fixed function f on the charge labels,

$$\frac{d}{dy}\mathbb{E}_p[f(q)] = -\operatorname{Cov}_p(f(q), q).$$

Proof. Differentiate the finite sum: $dW(e^y)/dy = -\sum_q q a_q e^{-qy} = -\mu W$. Logarithmic differentiation of p_q gives $\mu - q$. Multiplication by q and summation gives $\mu^2 - \mathbb{E}[q^2] = -V$. The same calculation with f gives the covariance identity. □

Theorem G3 — unique positive charge-neutral state. The finite Saha model has exactly one positive solution. Its electron density lies in $0 < x < NZ$, and its populations are the $p_q(x)$ above.

Proof. Let $F(x) = x - N\mu(x)$. It is continuously differentiable and

$$F'(x) = 1 + \frac{NV(x)}{x} > 1.$$

As $x \downarrow 0$, the highest-charge weight dominates, so $p_Z \to 1$, $\mu \to Z$, and $F \to -NZ < 0$. At $x = NZ$, strict positivity of lower charge populations gives $\mu(NZ) < Z$, hence $F(NZ) > 0$. The intermediate value theorem gives a root and strict monotonicity gives uniqueness. Ratios and normalization uniquely prescribe the populations for a given x; therefore the complete neutral solution is unique. □

Corollary G3a — stable log equation and polynomial form. The log-neutrality function

$$g(y) = y - \log N - \log \mu(e^y)$$

has derivative $g'(y) = 1 + V/\mu > 1$, tends to $-\infty$ as $y \to -\infty$, and is positive at $y = \log(NZ)$. It has the same unique root. Equivalently, x is the unique positive root of

$$P(x) = \sum_{q=0}^{Z} a_q x^{Z-q+1} - N \sum_{q=0}^{Z} q a_q x^{Z-q}.$$

The degree is $Z + 1$. This is a uniqueness theorem for the physical positive root, not a claim that all polynomial roots are real.

For helium, $Z = 2$, writing $u = x/N$, $\alpha = S_0/N$ and $\beta = S_1/N$, the independent equation is

$$u^3 + \alpha u^2 + (\alpha\beta - \alpha)u - 2\alpha\beta = 0, \qquad 0 < u < 2.$$

5.3 3. Exact comparative statics

All quantities below are evaluated at the unique neutral solution. Their smooth dependence follows from Theorem G3 and the implicit function theorem, since $F'(x) > 1$.

Theorem G4 — density response and stochastic ordering. At fixed positive Saha coefficients,

$$\frac{d\log x}{d\log N} = \frac{\mu}{\mu + V} \in (0, 1), \qquad \frac{d\mu}{d\log N} = -\frac{\mu V}{\mu + V} < 0,$$

and

$$\frac{d\log p_q}{d\log N} = (\mu - q)\frac{\mu}{\mu + V}.$$

Thus electron density increases with nucleus density, whereas the mean charge decreases. The neutral fraction increases. Every upper-charge tail $\Pr(q \geq r)$, $1 \leq r \leq Z$, strictly decreases with nucleus density. Intermediate individual charge populations need not be monotone over an entire density range.

Proof. Differentiate $x = N\mu(x)$:

$$\frac{dx}{dN} = \frac{\mu}{1 + NV/x} = \frac{\mu^2}{\mu + V} > 0.$$

Use $x = N\mu$ to obtain the log derivative and differentiate $\mu = x/N$ to obtain its derivative. Lemma G2 gives the formula for individual probabilities. For an upper-tail indicator $f(q) = \mathbf{1}_{q \geq r}$, the covariance in Lemma G2 is strictly positive. Indeed, for independent charge labels q, q',

$$\mathrm{Cov}(f(q), q) = \frac{1}{2}\sum_{i,j} p_i p_j (f(i) - f(j))(i - j) > 0,$$

because every term is nonnegative and at least one pair straddles the threshold. Since x increases with N, the tail decreases. Equivalently, for $x_2 > x_1$, the likelihood ratio $p_q(x_2)/p_q(x_1)$ is a positive constant times $(x_1/x_2)^q$, which decreases strictly with q. □

Theorem G5 — coefficient and parameter response. Let θ vary the positive coefficients smoothly at fixed N. Put

$$h_q = \partial_\theta \log a_q, \qquad C_h = \mathrm{Cov}_p(q, h_q).$$

Then

$$\partial_\theta \log x = \frac{C_h}{\mu + V}, \qquad \partial_\theta \mu = \frac{\mu C_h}{\mu + V},$$

and

$$\partial_\theta p_q = p_q \left[h_q - \mathbb{E}(h) - (q - \mu) \frac{C_h}{\mu + V} \right].$$

If h_q is increasing and nonconstant, then $C_h > 0$, and mean charge increases strictly. In particular, increasing any one coefficient S_r increases mean charge: for $\theta = \log S_r$, $h_q = \mathbf{1}_{q>r}$.

Proof. At fixed x, logarithmic differentiation gives $\partial_\theta p_q = p_q(h_q - \mathbb{E}h)$, so $\partial_\theta \mu = C_h$. The additional change through x contributes $-V\partial_\theta \log x$ by Lemma G2. Differentiating neutrality at fixed N gives $\partial_\theta \mu = \mu \partial_\theta \log x$, hence $(\mu + V)\partial_\theta \log x = C_h$. Substitution proves the displayed formulas. The covariance sign follows from the pairwise identity used in Theorem G4. □

For simultaneous variations of density and coefficients, the same computation gives the differential identity

$$d \log x = \frac{\mu}{\mu + V} \, d \log N + \frac{\mathrm{Cov}(q, d \log a_q)}{\mu + V},$$

$$d\mu = \frac{\mu}{\mu + V} \left[\mathrm{Cov}(q, d \log a_q) - V \, d \log N \right].$$

Here $d \log a_q$ is a charge-indexed differential; its covariance is taken over the current equilibrium populations.

5.4 4. Physical Saha coefficients and the VOTKP input

For the stated ideal, nonrelativistic, thermodynamic-equilibrium model,

$$S_q(T) = \frac{2}{\lambda_e(T)^3} \frac{U_{q+1}(T)}{U_q(T)} \exp[-I_q/(k_B T)], \qquad \lambda_e(T) = \frac{h}{\sqrt{2\pi m_e k_B T}}.$$

The factor 2 accounts for electron spin under this partition convention. U_q are positive internal partition weights and must use a consistent truncation and energy reference. The previous numerical helium experiment used ground-state weights $(1, 2, 1)$, not a complete excited-state equation of state.

Substituting Theorem G1 gives the exact re-expression

$$S_q(T) = \frac{2}{\lambda_e(T)^3} \frac{U_{q+1}(T)}{U_q(T)} \exp \left[-\frac{I_0}{k_B T} \prod_{j<q} \frac{1 + \Gamma_j}{1 - \Gamma_j} \right].$$

This is Saha with a reconstructed threshold input, not a new ionization law. It exposes the independent inputs: the absolute thermal energy ratio, density relative to the electron thermal wavelength, internal partition ratios, and a justified equilibrium assumption.

Theorem G6 — temperature response for fixed partition weights. Assume m_e, h, k_B fixed, every $I_q > 0$, and every $U_q > 0$ independent of T. Define cumulative removal energies $E_0 = 0$, $E_q = \sum_{j<q} I_j$. At fixed nucleus density,

$$\frac{d\log x}{d\log T} = \frac{\mathrm{Cov}\left(q, \frac{3}{2}q + \frac{E_q}{k_B T}\right)}{\mu + V} > 0, \qquad \frac{d\mu}{d\log T} > 0.$$

Proof. Telescoping the internal partition ratios gives $a_q = (2/\lambda_e^3)^q (U_q/U_0) e^{-E_q/(k_B T)}$. Its logarithmic temperature derivative is $h_q = 3q/2 + E_q/(k_B T)$. Each adjacent increment is $3/2 + I_q/(k_B T) > 0$. Theorem G5 applies. □

This theorem concerns electron density and mean charge. It does not assert monotonicity of every individual charge fraction. With temperature-dependent $U_q(T)$, the derivative includes $d\log(U_q/U_0)/d\log T$; the stated positivity proof then needs an additional condition. It does not follow solely from threshold positivity.

At fixed T, N, U, m_e, increasing any threshold I_r decreases mean charge. In fact Theorem G5 gives

$$\frac{\partial \mu}{\partial I_r} = -\frac{\mu}{k_B T(\mu + V)} \mathrm{Cov}(q, \mathbf{1}_{q>r}) < 0.$$

Consequently, common multiplicative scaling of all threshold energies by a factor greater than one at fixed specimen conditions leaves every Γ_q fixed but changes the neutral solution. The missing absolute anchor is physically consequential.

Theorem G7 — complete scaling covariance. In the abstract finite Saha model, the simultaneous transformation $S_q \mapsto cS_q$, $N \mapsto cN$, $c > 0$, preserves every charge fraction and takes $x \mapsto cx$.

Proof. The ratios S_q/x and neutrality $x = N\mu$ are preserved by the proposed transformation. Theorem G3 makes the transformed solution unique. Equivalently, the fractions depend only on the dimensionless coefficients S_q/N. □

For the physical coefficients, take positive scale factors a, b and impose

$$m_e' = a m_e, \quad T' = bT, \quad I_q' = bI_q, \quad \frac{U_{q+1}'(T')}{U_q'(T')} = \frac{U_{q+1}(T)}{U_q(T)}, \quad N' = (ab)^{3/2} N.$$

Then $S_q' = (ab)^{3/2} S_q$, so all populations are invariant. Fixed electron mass gives the density factor $b^{3/2}$. Scaling mass and energy/temperature together by c gives the density factor c^3. These are conditional covariances of a specified model. They are not an assertion that fundamental constants or real helium thresholds can be changed arbitrarily, nor a cosmological mechanism.

Corollary G7a — record insufficiency in the declared model. No function of the complete anchored atomic threshold record alone determines mean charge for all positive specimen densities at fixed T, U, m_e: Theorem G4 makes that mean strictly density-dependent. No function of the unanchored ratio vector alone determines it even at fixed T, N, U, m_e: common threshold scaling leaves that vector fixed and changes the mean by Theorem G6's threshold derivative. These are exact non-identifiability statements, not empirical regression findings.

5.5 5. Retained and hidden amplitudes: the full reduced equation

Let $\mathcal{H} = \mathcal{H}_A \oplus \mathcal{H}_B$ be finite-dimensional complex inner-product spaces with a specified orthogonal split. Let

$$H = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix} = H^\dagger, \qquad \psi(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \quad i\hbar\dot{\psi} = H\psi,$$

with $\hbar > 0$. The partition is physical data. If it is an invariant spectral subspace of this same undriven H, then $B = 0$ and the coupling kernel is trivial.

Theorem R1 — exact reduced equation, including hidden preparation. Define

$$K(t) = Be^{-iDt/\hbar}B^\dagger, \qquad J(t) = Be^{-iDt/\hbar}y(0).$$

Then the retained amplitude solves

$$\dot{x}(t) = -\frac{i}{\hbar}Ax(t) - \frac{i}{\hbar}J(t) - \frac{1}{\hbar^2}\int_0^t K(t-s)x(s)\,ds.$$

Proof. Variation of constants in the hidden equation gives

$$y(t) = e^{-iDt/\hbar}y(0) - \frac{i}{\hbar}\int_0^t e^{-iD(t-s)/\hbar}B^\dagger x(s)\,ds.$$

Substituting into $\dot{x} = -i(Ax + By)/\hbar$ gives the formula, including the minus sign in front of the integral. □

The specified $A, K, J, x(0)$ uniquely determine the finite-dimensional retained evolution. This can be seen directly from the original finite linear system, or by the uniqueness theorem for a linear Volterra equation with continuous kernel. A kernel without the hidden-source term does not specify evolution when $y(0) \neq 0$.

Theorem R2 — positive channel measure and Schur response. If E_D is the spectral projection measure of D, the finite positive operator-valued measure

$$d\mathsf{M}(\lambda) = B\,dE_D(\lambda)\,B^\dagger$$

satisfies

$$K(t) = \int e^{-i\lambda t/\hbar}\,d\mathsf{M}(\lambda), \qquad \Sigma(z) = B(zI - D)^{-1}B^\dagger = \int \frac{d\mathsf{M}(\lambda)}{z - \lambda}.$$

When the indicated inverses exist, the retained block of $(zI - H)^{-1}$ is

$$G_A(z) = \big(zI - A - \Sigma(z)\big)^{-1}.$$

In particular, $K(0) = \mathsf{M}(\mathbb{R}) = BB^\dagger$ is only the total channel mass, not the measure's frequency distribution.

Proof. The spectral theorem yields the two integral formulas, and each atom of the measure is positive semidefinite because $BE_D(\{\lambda\})B^\dagger \succeq 0$. Block Gaussian elimination gives the Schur inverse formula. □

For $\operatorname{Im} z > 0$,

$$\operatorname{Im}\Sigma(z) = -(\operatorname{Im} z)B\big((\operatorname{Re} z - D)^2 + (\operatorname{Im} z)^2 I\big)^{-1} B^\dagger \preceq 0.$$

Also K is positive definite as an operator-valued function: for any finite times t_j and retained vectors u_j,

$$\sum_{j,k} u_j^\dagger K(t_j - t_k) u_k = \left\| \sum_k e^{iDt_k/\hbar} B^\dagger u_k \right\|^2 \geq 0.$$

These properties constrain admissible channel kernels. They do not turn a scalar delay curve into a fully reconstructed matrix measure.

5.6 6. Equal full spectra and coupling footprint, unequal later response

The following exact witness is deliberately finite-dimensional. It proves an information limitation for spectral lists and instantaneous coupling footprints, not the existence of a particular gas preparation in a laboratory.

Set $\hbar = 1$, fix the retained subspace $\operatorname{span}\{e_0\} \subset \mathbb{R}^4$, and let

$$\Lambda = \operatorname{diag}(-3, -1, 1, 3), \qquad w^{(0)} = (1,1,1,1)/4, \quad w^{(1)} = (3,7,1,5)/16.$$

For either positive unit-sum weight vector w, set $s = (\sqrt{w_0}, \dots, \sqrt{w_3})^T$, $v = e_0 - s$, and define the explicit Householder matrix

$$R_w = I - \frac{vv^T}{1 - s_0}, \qquad H_w = R_w \Lambda R_w^T.$$

The denominator is nonzero for these weights. Since $v^T v = 2(1 - s_0)$, one has $R_w^T R_w = I$, $R_w^T = R_w$, and $R_w e_0 = s$. Thus H_w is a real symmetric Hamiltonian with the declared eigenvalues and a fixed retained projector $e_0 e_0^T$. This is a constructive matrix formula, not merely an existence appeal to a change of basis.

Theorem R3 — spectral and instantaneous-footprint insufficiency. Both $H_{w^{(0)}}$ and $H_{w^{(1)}}$ have exactly the spectrum $\{-3, -1, 1, 3\}$, retained diagonal $A = 0$, and $BB^T = 5$. With the same initial full state $\psi(0) = e_0$, hence $x(0) = 1$ and $y(0) = 0$, their retained survival probabilities at $t = \pi/4$ are respectively 0 and $1/32$.

Proof. The unit-sum moments satisfy

$$\sum_j w_j^{(0)} \lambda_j = \sum_j w_j^{(1)} \lambda_j = 0, \qquad \sum_j w_j^{(0)} \lambda_j^2 = \sum_j w_j^{(1)} \lambda_j^2 = 5.$$

These give $A = e_0^T H_w e_0 = 0$ and $BB^T = e_0^T H_w^2 e_0 - A^2 = 5$. The retained amplitude is

$$a_w(t) = e_0^T e^{-iH_w t} e_0 = \sum_j w_j e^{-i\lambda_j t}.$$

For $w^{(0)}$, $a_0(t) = (\cos 3t + \cos t)/2$. The difference weights are $(-1, 3, -3, 1)/16$, so

$$a_1(t) - a_0(t) = \frac{i}{8}(3\sin t - \sin 3t) = \frac{i}{2}\sin^3 t.$$

At $t = \pi/4$, $a_0 = 0$ and $a_1 = i\sqrt{2}/8$, whose squared magnitudes are 0 and $1/32$. Both hidden-source terms are zero because the same initial state is wholly retained. □

The weighted third moments are 0 and 3. This identifies the first lost distinction: BDB^T is respectively 0 and 3, and $K'(0) = -iBDB^T$ differs even though $K(0)$ agrees.

The Schur responses are also explicit. With

$$Q(z) = (z^2 - 1)(z^2 - 9) = z^4 - 10z^2 + 9,$$

the retained resolvents are

$$G_0(z) = \frac{z^3 - 5z}{Q(z)}, \qquad G_1(z) = \frac{z^3 - 5z + 3}{Q(z)}.$$

Since $A = 0$, $\Sigma = z - G^{-1}$, hence

$$\Sigma_0(z) = \frac{5z^2 - 9}{z^3 - 5z}, \qquad \Sigma_1(z) = \frac{5z^2 + 3z - 9}{z^3 - 5z + 3}.$$

These meromorphic identities hold away from their poles, with analytic continuation where removable. Both self-energies begin with $5/z$ at large z, but their next coefficients differ. The full eigenvalue list supplies possible energies; their spectral weights relative to the fixed retained observable carry additional information.

The same finite-difference null vector appears in the companion classical moment-matching construction. The shared fact is cancellation of the first three moments. Classical streaming density, quantum survival probability and an atomic removal-threshold list remain different observables and must not be equated.

5.7 7. Same kernel, different hidden initial source

Proposition R4 — kernel-only evolution is underdetermined. Take

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \psi_\pm(0) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}, \quad \hbar = 1.$$

The retained amplitudes at time zero are identical, $A = D = 0$, and $K(t) = 1$ is identical. Nevertheless

$$|x_\pm(t)|^2 = \frac{1}{2}(1 \pm \sin 2t),$$

so the retained probabilities at $t = \pi/4$ are 1 and 0.

Proof. Here $e^{-iHt} = \cos t\, I - i \sin t\, H$. Its first component gives $x_\pm = (\cos t \pm \sin t)/\sqrt{2}$. Squaring gives the result. The hidden sources are $J_\pm = \pm i/\sqrt{2}$, so Theorem R1 predicts the distinction exactly. □

This example changes the initial state, whereas Theorem R3 keeps the initial state fixed and changes its resolved spectral weights through the Hamiltonian. They demonstrate two separate

omissions. A proposed history experiment must declare which one it varies and which current measurements it matches.

5.8 8. Evidence and formalization boundary

The incoming matter-disk §8 reverse-recovery construction is developed in `gas_peel_inverse.md`, which supplies Theorems G8–G9: population-combination reconstruction of the threshold ladder, exact partition corrections, and deterministic inversion-error bounds. It is part of this account. Its synthetic excited-partition opposite was specified in a separate prospective addendum before execution. The retained-response theorem labels were changed from M1–M4 to R1–R4 to avoid a collision with the manuscript's separate moment theorem; this administrative relabeling changes no claim, tolerance or test prediction.

Theorem labels G1–G7 and R1–R3, with their corollaries and Proposition R4, are human-readable analytic proofs in this supplement. Machine-checked coverage is only what the project's Lean theorem manifest explicitly lists; a passing numerical script is not a Lean proof. No theorem here assumes a desired physical conclusion under a renamed hypothesis.

`verify_formal_gas.py` executes the prospective checks specified in `PREREGISTERED_FORMAL_CONTROLS.md`, produces `formal_gas_results.json`, and records the protocol SHA-256. Rational ladder reconstruction, manufactured neutrality polynomials and resolvent numerators are checked in exact arithmetic. Finite-difference responses and matrix exponentials are floating-point implementation checks with recorded tolerances. No threshold, temperature or coefficient is fitted to measured populations or delays.

The original helium source grades remain E for 24.587389011 eV and T for 54.4177655282 eV. The archival xenon value remains a retrospective published channel-delay contrast, 114.323 ± 35.0164 as at 75.36 eV, not a matched-history gas experiment or a recovered positive matrix measure. Its raw-scan reconstruction, systematic-error audit and microscopic kernel inference remain separate work.

The next physical test should specify a real specimen, retain its measured preparation variables, choose a justified equilibrium or kinetic model, and predict a new resolved response before inspecting it. The finite proofs above determine which records are sufficient inside their declared models and which proposed summaries necessarily lose information.

6 Reverse recovery: a gas carries the peel under a specified closure

Supplement G8–G9 to `formal_gas_response.md`, 3 November 2026. The newly supplied `matter-disk` branch, commit `881386a0ab00d917af6334a01423fb8b22469f5e`, adds an inverse construction in §8. It is consistent with the forward insufficiency result: the inverse uses population and specimen information that the atomic record alone does not supply. This section retains that construction and proves its exact general form.

Let $\beta = 1/(k_B T)$, with energies in the matching units. Suppose all populations are positive and satisfy the Saha ratios at the sampled temperatures. Electron density, nucleus density and temperature can vary between samples; every state-dependent correction must use its own sample's values.

Theorem G8 — population combinations recover the threshold record. For $0 \le q < Z-1$,

$$L_q = \log \frac{n_q n_{q+2}}{n_{q+1}^2} = \log \frac{U_q U_{q+2}}{U_{q+1}^2} - \beta(I_{q+1} - I_q).$$

The full-ladder combination satisfies

$$L_B = \log \frac{n_0}{n_Z\,(n_e \lambda_e^3)^Z} = \log \frac{U_0}{2^Z U_Z} + \beta B, \qquad B = \sum_{q=0}^{Z-1} I_q.$$

The L_q are population-only ratios and cancel electron density and thermal wavelength algebraically. L_B is a **corrected** ratio: evaluating it requires the sample's $n_e \lambda_e^3$. Its right side has no remaining explicit density factor, but it is not an observable constructed from charge fractions alone.

Proof. Dividing two adjacent Saha relations gives $n_q n_{q+2}/n_{q+1}^2 = S_{q+1}/S_q$. Their common electron-density and wavelength factors cancel. Multiplying all Z Saha relations gives

$$\frac{n_Z}{n_0} = \frac{2^Z}{(n_e \lambda_e^3)^Z} \frac{U_Z}{U_0} e^{-\beta B}.$$

Taking logarithms proves both identities. Since $n_q = N p_q$, the common nucleus density also cancels from the displayed ratios. No charge-neutrality assumption is needed for these identities themselves; it is needed only if n_e is inferred from pure-species populations rather than supplied independently. □

If the partition weights and thresholds are independent of temperature, take two distinct inverse temperatures. The slopes give

$$D_q = -\frac{\Delta L_q}{\Delta \beta} = I_{q+1} - I_q, \qquad B = \frac{\Delta L_B}{\Delta \beta}.$$

The complete threshold ladder follows from

$$I_0 = \frac{1}{Z}\left[B - \sum_{r=0}^{Z-2} (Z-1-r) D_r \right], \qquad I_q = I_0 + \sum_{r<q} D_r.$$

Indeed, summing $I_q = I_0 + \sum_{r<q} D_r$ counts each D_r exactly $Z-1-r$ times. This reconstruction includes the absolute scale; the Cayley vector follows by Theorem G1. With known partition

functions and known $n_e\lambda_e^3$, one temperature already gives the individual thresholds directly from each adjacent Saha relation. The two-temperature construction instead removes constant partition intercepts.

For helium with $U = (1, 2, 1)$, write $D = I_1 - I_0$. Then

$$\log \frac{n_0 n_2}{n_1^2} = -\beta D - \log 4, \qquad \log \frac{n_0}{n_2 n_e^2 \lambda_e^6} = \beta B - \log 4, \qquad \Gamma_0 = D/B.$$

These are precisely the incoming §8 identities. Generating Saha populations from an input ladder and reading that ladder back verifies a model identity and the code, not an independent experimental determination of atomic thresholds.

Theorem G9 — partition correction and the excited-state opposite. If positive $U_q(T)$ vary with temperature but are known, define

$$\widetilde{L}_q = L_q - \log \frac{U_q U_{q+2}}{U_{q+1}^2}, \qquad \widetilde{L}_B = L_B - \log \frac{U_0}{2^Z U_Z}.$$

For temperature-independent thresholds, their two-temperature slopes recover D_q and B exactly. Omitting the correction gives biases

$$\widehat{D}_q = D_q - \frac{\Delta \log(U_q U_{q+2}/U_{q+1}^2)}{\Delta\beta}, \qquad \widehat{B} = B + \frac{\Delta \log(U_0/(2^Z U_Z))}{\Delta\beta}.$$

Proof. Subtract the known partition terms in Theorem G8 and take finite differences in β. □

For finite electronic excitation partitions $U_q(\beta) = \sum_j g_{qj} e^{-\beta\epsilon_{qj}}$, with fixed positive weights g_{qj} and fixed nonnegative excitation energies, let $\bar{\epsilon}_q$ be their normalized thermal mean. Differentiation gives $d \log U_q / d\beta = -\bar{\epsilon}_q$. Thus the **local**, uncorrected slope readings for helium are

$$D_{\text{app}} = D + \bar{\epsilon}_0 + \bar{\epsilon}_2 - 2\bar{\epsilon}_1, \qquad B_{\text{app}} = B - \bar{\epsilon}_0 + \bar{\epsilon}_2.$$

For a finite temperature interval these terms are interval averages in inverse temperature, not evaluations at an arbitrary midpoint. Consequently the naive quotient $D_{\text{app}}/B_{\text{app}}$ generally differs from the atomic Γ_0.

The registered opposite uses $U_0 = U_2 = 1$ and the synthetic excited partition $U_1(T) = 2 + 4e^{-10\,\text{eV}/(k_B T)}$. It is not an asserted helium level scheme. Its purpose is to demonstrate a controlled failure of the constant-partition inference and exact recovery after the known partition correction. Actual helium excited-state partitions, continuum lowering, non-LTE populations and experimental uncertainty remain separate physical validation tasks.

Stability of slope inversion. If each corrected logarithmic combination has endpoint errors bounded by η_1, η_2, its two-temperature slope error is at most $(\eta_1 + \eta_2)/|\Delta\beta|$. Thus close temperatures amplify observational errors even though the exact identity holds. For helium, suppose $|\widehat{D} - D| \le \eta_D$, $|\widehat{B} - B| \le \eta_B < B$, and $B > 0$. Then

$$\left| \frac{\widehat{D}}{\widehat{B}} - \frac{D}{B} \right| \le \frac{\eta_D + |D/B|\eta_B}{B - \eta_B}.$$

This follows by writing the numerator as $B(\widehat{D} - D) - D(\widehat{B} - B)$ and using $|\widehat{B}| \ge B - \eta_B$. No statistical independence is assumed in this deterministic error bound. Vanishing or unmeasurably rare

populations make the logarithmic inversion undefined or poorly conditioned; numerical closure tests do not remove that measurement limitation.

6.1 Machine and numerical coverage

Theorem G8 and G9 have explicit analytic proofs above. The prospective numerical checks are in `PREREGISTRATION_ADDENDUM_GAS_PEEL.md`, with separate status in `formal_gas_results.json`. Lean coverage is only what the final theorem manifest lists. The original §8 reproduction remains a retrospective verification of supplied results; the synthetic partition opposite was specified before executing the new control.

7 Finite records and later responses

The gas and retained-channel constructions below share a precise piece of algebra: finitely many moments of a positive measure do not determine its full response transform. This is a relation between mathematical representations. The measured quantities and physical hypotheses in the two applications remain different.

7.1 A constructive finite-moment theorem

Theorem M1 (positive measures with a matched finite record). Let $m \geq 1$, $a \in \mathbb{R}$, $h \in \mathbb{R}\setminus\{0\}$, and $x_j = a + jh$ for $0 \leq j \leq m$. Let $p_j > 0$ and $\sum_j p_j = 1$. Define

$$w_j = (-1)^j \binom{m}{j}, \qquad q_j = p_j + \delta w_j,$$

where $0 < |\delta| < \min_j p_j/|w_j|$. Then p and q are distinct, strictly positive probability vectors on the same support. Their moments of every order $r < m$ agree, whereas

$$\sum_{j=0}^{m} (q_j - p_j) x_j^m = \delta (-1)^m m! h^m \neq 0.$$

For the response transform $F_p(t) = \sum_j p_j e^{-itx_j}$,

$$F_q(t) - F_p(t) = \delta e^{-iat}(1 - e^{-iht})^m.$$

Consequently the complex responses differ at every real t for which $ht \notin 2\pi\mathbb{Z}$, and agree at the recurrence times. All derivatives through order $m-1$ agree at $t = 0$, but the order-m derivatives do not.

Proof. The binomial identity gives $\sum_j w_j = (1-1)^m = 0$. The bound on δ makes every q_j positive, hence it also normalizes to one. For a polynomial f, the mth forward difference satisfies

$$\Delta_h^m f(a) = \sum_{j=0}^{m} (-1)^{m-j} \binom{m}{j} f(a + jh).$$

Each application of Δ_h lowers the degree of a nonconstant polynomial by one. Thus this expression is zero for $f(x) = x^r$, $r < m$, and equals $m!h^m$ for $f(x) = x^m$. Multiplication by $(-1)^m\delta$ proves the moment statements. Applying the ordinary binomial theorem to the exponential sum yields

$$\sum_j \delta (-1)^j \binom{m}{j} e^{-it(a+jh)} = \delta e^{-iat} \sum_j \binom{m}{j} (-e^{-iht})^j = \delta e^{-iat}(1 - e^{-iht})^m.$$

Its zeros on the real axis are exactly the stated recurrence times. Differentiating the finite sums is legitimate term by term and gives the derivative claims. □

Corollary M2 (no universal response from a finite moment record). No single-valued rule of the first m moments, including normalization, can reproduce the whole response transform for all strictly positive probability measures on this support. If such a rule existed, the equal inputs furnished by M1 would give equal responses, contradicting its explicit formula. The conclusion concerns this restricted record, not the complete weight vector.

The anchored VOTKP ladder has a different status: it is an invertible coordinate representation of the complete threshold list. M2 does not assert information loss in that invertible map. The missing response information is the preparation, channel weights, matrix elements or additional state variables which the threshold list was never claimed to determine.

7.2 Readout dependence is part of the theorem

Equality of a few derivatives and equality of an entire response are distinct statements. Likewise, a complex-amplitude separation need not survive projection to an intensity. On a symmetric support, if two probability measures are reflections of each other, then their characteristic functions are complex conjugates on the real axis and have the same squared magnitude. Thus an intensity-only detector can erase a distinction that a phase-resolved readout retains.

For $m = 3$, use support $(-3, -1, 1, 3)$. The uniform weights and $(3, 7, 1, 5)/16$ have equal normalization, mean0 and second moment5. Their third moments are0 and3. At $t = \pi/4$, their response intensities are0 and$1/32$. The probability vectors $(3, 7, 1, 5)/16$ and $(5, 1, 7, 3)/16$, however, are reflections; they have different complex responses but equal intensities at every real time. Both the separating witness and the nonseparating control are retained.

This mechanism is a standard finite-difference/moment construction. Its use here supplies an explicit audit of which record and readout keep a distinction. It is not asserted to be a new theorem in the moment problem or an observation of memory in a laboratory gas.

8 An exact gas-transport witness with matching density, flow and stress

Prepared for Jeromie Beasley's matter–gas–VOTKP programme, 3 November 2026. These are fully specified analytic models and proofs, not new laboratory observations. The finite-difference moment-null identity is developed more generally in the manuscript's M1; the construction below specializes it to a positive velocity distribution and a physical transport equation.

8.1 K1. Two positive preparations with the same first three moments — proved here

Let x be periodic with period L, choose $k = 2\pi m/L$ for an integer $m \geq 1$, and fix

$$\rho_* > 0, \quad 0 < \eta < 1, \quad v_0 > 0, \quad 0 < a < 1/12, \quad \sigma \geq 0.$$

Put

$$u = (-3, -1, 1, 3), \qquad d = (-1, 3, -3, 1), \qquad \rho(x) = \rho_*[1 + \eta\cos(kx)].$$

For $\sigma > 0$, let g_σ be the centered normalized Gaussian of variance σ^2. At $\sigma = 0$, interpret $g_0(v - v_j)\,dv$ as the point mass at v_j. Define

$$p_\pm(v) = \sum_{j=0}^{3} (\tfrac{1}{4} \pm a d_j) g_\sigma(v - v_0 u_j), \qquad f_\pm(x, v, 0) = \rho(x) p_\pm(v).$$

Both are nonnegative normalized conditional velocity distributions, strictly positive for $\sigma > 0$. Their pointwise initial moments are

$$M_\ell^\pm(x, 0) = \int_{\mathbb{R}} v^\ell f_\pm(x, v, 0)\,dv,$$

$$M_0^+ = M_0^- = \rho(x), \qquad M_1^+ = M_1^- = 0, \qquad M_2^+ = M_2^- = (5v_0^2 + \sigma^2)\rho(x).$$

Their third moments differ:

$$M_3^\pm(x, 0) = \pm 48 a v_0^3 \rho(x), \qquad \Delta M_3(x, 0) = 96 a v_0^3 \rho(x),$$

where $\Delta = (+) - (-)$.

Proof. The four weights sum to one because $\sum_j d_j = 0$. Their smallest possible value is $1/4 - 3a > 0$. Direct exact sums give

$$\sum_j d_j u_j^\ell = 0 \quad (\ell = 0, 1, 2), \qquad \sum_j d_j u_j^3 = 48,$$

while $\frac{1}{4}\sum_j u_j = 0$, $\frac{1}{4}\sum_j u_j^2 = 5$, and $\sum_j u_j^3 = 0$. Convolution with the centered Gaussian adds σ^2 to each second moment and $3\sigma^2 v_0 u_j$ to each third moment. The corresponding perturbation sums still vanish at orders at most two and the additional third-moment term vanishes because $\sum_j d_j u_j = 0$. □

Thus number density, mean velocity and kinetic stress are equal **pointwise**, not merely after spatial averaging. With particle mass m_p, the kinetic-energy density is also equal, $m_p M_2/2$. This is a one-dimensional illustration, so it is not by itself a claim about every component of a three-dimensional stress tensor.

8.2 K2. Same transport law, different later density — proved here

Evolve both preparations by the same force-free collisionless Boltzmann equation,

$$\partial_t f + v\partial_x f = 0, \qquad f_\pm(x,v,t) = f_\pm(x - vt, v, 0).$$

Let $n_\pm(x,t) = \int f_\pm(x,v,t)\,dv$. With $q = kv_0 t$, the exact densities are

$$n_\pm(x,t) = \rho_* \left[1 + \eta e^{-\sigma^2 k^2 t^2/2} \left(\tfrac{1}{2}[\cos q + \cos(3q)]\cos(kx) \mp 8a \sin^3 q \sin(kx)\right)\right].$$

In particular,

$$\boxed{\Delta n(x,t) = -16a\eta\rho_* e^{-\sigma^2 k^2 t^2/2} \sin(kx)\sin^3(kv_0 t).}$$

Proof. Along a characteristic, $x - vt$ is constant, proving the solution formula. The Gaussian Fourier transform gives

$$\int g_\sigma(v - v_0 u_j) e^{-ikvt}\,dv = e^{-iku_j v_0 t} e^{-\sigma^2 k^2 t^2/2}.$$

The unperturbed symmetric weights contribute $[\cos q + \cos(3q)]/2$. For the perturbation,

$$\sum_j d_j \cos(qu_j) = 0, \qquad \sum_j d_j \sin(qu_j) = 2\sin(3q) - 6\sin q = -8\sin^3 q.$$

Expanding $\cos(kx - kvt)$ and combining these identities proves both formulas. □

For the exact discrete-velocity instance

$$a = 1/24, \quad \eta = 1/2, \quad \rho_* = v_0 = k = 1, \quad \sigma = 0, \quad x = t = \pi/2,$$

the two later densities are

$$n_+ = 5/6, \qquad n_- = 7/6, \qquad \Delta n = -1/3.$$

For $\sigma > 0$, the distributions are smooth and positive and the separation remains nonzero at the same event, reduced by $e^{-\sigma^2\pi^2/8}$. The discrete case is therefore not essential to the qualitative separation.

8.3 K3. What the missing moment does — proved here

The transport equation implies the hierarchy

$$\partial_t M_\ell + \partial_x M_{\ell+1} = 0.$$

For the two preparations in K1,

$$\Delta n(x,0) = \partial_t \Delta n(x,0) = \partial_t^2 \Delta n(x,0) = 0,$$

but

$$\partial_t^3 \Delta n(x,0) = -\partial_x^3 \Delta M_3(x,0) = -96 a\eta\rho_* k^3 v_0^3 \sin(kx).$$

Proof. Integrate the kinetic equation against v^ℓ; the finite mixture with Gaussian tails justifies differentiation and integration. The discrete measure case is the same finite sum. Iterate the hierarchy three times and use K1. Equivalently, differentiate the exact formula in K2. □

The stress separates first through its derivative, the flow next, and the density at third order. The full kinetic state obeys a Markovian evolution. The history dependence appears only after retaining the three lower moments and discarding the higher velocity structure. These are matching **specified present measurements**, not the same complete physical state. No contradiction with deterministic dynamics is involved.

8.4 K4. A precise erasure intervention before release — proved here

Define the common symmetric velocity target

$$p_0(v) = \tfrac{1}{4}\sum_{j=0}^{3} g_\sigma(v - v_0 u_j).$$

During an explicitly separate preparation interval of duration s, hold the spatial profile fixed and apply the velocity-reset map

$$\mathcal{E}_s[f](x,v) = e^{-\nu s} f(x,v) + (1 - e^{-\nu s})\rho(x) p_0(v), \qquad \nu \geq 0.$$

For the K1 preparations this map preserves their common density, flow and stress, preserves positivity, and multiplies their difference by $e^{-\nu s}$. If free streaming starts **after** this preparation interval, then the density contrast in K2 is multiplied by exactly $e^{-\nu s}$.

Proof. The map is a convex combination of positive distributions with the same first three moments. Its difference on the two preparations is $e^{-\nu s}(f_+ - f_-)$. Free streaming and integration in velocity are linear, so this factor carries through to the later density contrast. □

The map can be implemented mathematically as a Poisson reset to a prescribed common velocity distribution while the spatial profile is held. It is not asserted to be the self-consistent collision operator of an isolated gas. If collisions and streaming act simultaneously with a moment-dependent local equilibrium, the two targets need not stay equal once the lower moments separate; the simple exponential factor has then **not** been proved. The separation of preparation and release is an essential hypothesis.

An early perturbation exposed to the reset for longer has a smaller later signal; a perturbation inserted immediately before release has less time to be erased. This gives a controlled mathematical version of the early/late distinction in the song without making a universal dense-gas claim.

8.5 Null tests and limitations

Control	Exact prediction	Interpretation
$a = 0$	$f_+ = f_-$, hence $\Delta n = 0$	Remove the hidden distinction
$\eta = 0$	$\Delta n = 0$ for all times	Homogeneous density is insensitive to the different velocity record
$\sin(kx) = 0$	Density contrast zero at that readout	Spatial placement matters
$kv_0 t \in \pi\mathbb{Z}$	Density contrast zero at that time	A null readout is not evidence that the complete distributions agree
Pre-release reset $s \to \infty$, $\nu > 0$	$\Delta n \to 0$	An intervention erases the prepared distinction
Include M_3 among retained observations	Preparations differ already initially	The original three-moment match is no longer the stated experiment
$\sigma > 0$	Exact Gaussian dephasing envelope	Smooth velocity spread reduces this density readout without making the full states identical

No gas pressure law, collision cross section, VOTKP ionization threshold, quantum memory kernel or laboratory calibration is inferred from this construction. It establishes a bounded mathematical claim: a density–flow–stress record alone does not determine later force-free transport for arbitrary positive velocity distributions. Testing whether an actual gas can prepare and retain this distinction requires a separate physical model and experiment.

9 What the Petermann factor establishes

The incoming disk work identifies a useful exact band average and a separate numerical relation between that average and a filtered block-spectrum exit. This section proves the first statement and preserves the second as an open conjectural identification. The distinction matters even when numerical agreement is excellent.

9.1 P1. A two-level projector formula

Let $H = d \cdot \sigma$, with $d \in \mathbb{C}^3$, and choose $E \neq 0$ satisfying $E^2 = d \cdot d$. The dot product in this equation is bilinear, without complex conjugation. Then

$$P_{\pm} = \frac{1}{2}\left(I \pm \frac{H}{E}\right)$$

are rank-one idempotent spectral projectors. Define their Petermann factor here by $K_{\pm} = \mathrm{Tr}(P_{\pm}^{*}P_{\pm})$, the squared Frobenius norm. Then

$$K_{+} = K_{-} = \frac{1}{2}\left(1 + \frac{\sum_{j=1}^{3}|d_j|^2}{|E|^2}\right) \geq 1.$$

Proof. The Pauli identities give $H^2 = (d \cdot d)I = E^2 I$ and $\mathrm{Tr}\, H = 0$, so $P_{\pm}^2 = P_{\pm}$ and $\mathrm{Tr}\, P_{\pm} = 1$. Moreover $\mathrm{Tr}(H^{*}H) = 2\sum_j |d_j|^2$. Expanding the norm and using the vanishing traces proves the formula. Finally $|E|^2 = |\sum_j d_j^2| \leq \sum_j |d_j|^2$. □

The definition is tied to the declared inner product. In a general nonorthogonal eigenbasis, left/right normalizations must be treated consistently. At an exceptional point with $E = 0$, this projector formula is singular and is not an admissible substitution into the finite expression.

9.2 P2. The exact average for the unbroken SSH band

Let $v, w, u \in \mathbb{R}$, $q(k) = v + we^{ik}$, and

$$H(k) = \begin{pmatrix} iu & \overline{q(k)} \\ q(k) & -iu \end{pmatrix}.$$

Assume the whole band is strictly in the real, nonzero-energy regime:

$$A = v^2 + w^2 - u^2 > |2vw| = |B|.$$

Choose $E(k) = \sqrt{A + B\cos k} > 0$. Then

$$K(k) = 1 + \frac{u^2}{E(k)^2}, \qquad \overline{K} = \frac{1}{2\pi}\int_0^{2\pi} K(k)\,dk = 1 + \frac{u^2}{\sqrt{A^2 - B^2}} = 1 + \frac{u^2}{E(0)E(\pi)}.$$

Proof. In P1, the Pauli vector is $(\mathrm{Re}\, q, \mathrm{Im}\, q, iu)$. Thus $|E|^2 = |q|^2 - u^2$ and $\sum |d_j|^2 = |q|^2 + u^2$, which gives the pointwise formula. With $A > |B|$, the substitution $t = \tan(k/2)$ on the two half-intervals gives

$$\int_0^{2\pi} \frac{dk}{A + B\cos k} = 2\int_{-\infty}^{\infty} \frac{dt}{(A+B) + (A-B)t^2} = \frac{2\pi}{\sqrt{A^2 - B^2}}.$$

The hypotheses make both factors positive, so $\sqrt{A^2 - B^2} = \sqrt{A+B}\sqrt{A-B} = E(0)E(\pi)$. □

It follows exactly that $\overline{K} = 2$ if and only if $E(0)E(\pi) = u^2$ within this domain. This equivalence concerns the **band average**. It does not prove that a finite block covariance crosses a chosen complex-coordinate boundary at that same parameter. Such an assertion must specify the block, boundary conditions, eigenvalue filter, numerical error and limiting procedure.

9.3 P3. A partition counterexample to an unrestricted exit equivalence

Set $w = 0$, choose $u \neq 0$ and $v^2 = 2u^2$, and partition the lattice into whole unit cells. Then the band is unbroken: $E^2 = v^2 - u^2 = u^2 > 0$, and P2 gives $\overline{K} = 2$. Nevertheless, no compressed projector eigenvalue leaves $\{0, 1\}$.

Proof. With $w = 0$, $H(k)$ is independent of k. Its real-space spectral projector is a direct sum of identical two-by-two cell projectors. Compressing to any union of whole cells leaves a direct sum of idempotents, so the compressed matrix is itself idempotent and has only the eigenvalues zero and one. Thus $\Gamma = 1 - 2\nu$ remains at ± 1, and a filter that removes both endpoint eigenvalues has an empty retained spectrum. As u varies while $u^2 < v^2$, this statement continues to hold, including the point $\overline{K} = 2$. There is no outward crossing of the unit circle. □

This is an exact counterexample to a universal claim over all such PT-symmetric bands and spatial partitions. It does not refute the finite numerical observation for the specified coupled family with $w = 1$. A proposed exit theorem must include its coupling and partition assumptions, rather than infer the exit from the average alone.

9.4 A coordinate is not a universal phase classifier

For a complex ratio $z \neq -1$, the Cayley map obeys $|(z-1)/(z+1)| < 1$ exactly when $\mathrm{Re}\, z > 0$. For example $z = 1 - i\gamma$, $\gamma > 0$, is complex and remains inside the disk. An unstable resonance width therefore does not, by itself, force the coordinate outside. A specified ratio of specified physical quantities is required. The incoming particle-side sweep is preserved as a research proposal; its headline inventory is not a set of verified new particle measurements.

Removing an arbitrary energy origin is also essential. The eigenvalues $(1, 2)$ have ratio 2 and Cayley coordinate $1/3$; subtracting $5/4$ from both gives $(-1/4, 3/4)$, ratio -3 and coordinate 2, although their energy difference stays 1. This does not invalidate VOTKP's removal thresholds, which are energy differences with a physical reference. It shows why a raw eigenenergy ratio is not automatically the same invariant.

Finally, a literal spectral census is a nonnegative integer. A negative coefficient extracted from a biorthogonal Rényi entropy is a property of that nonpositive-state entropy convention; it is not automatically a negative number of boundary channels. If a nonnegative census had $N(L) = a \log L + O(1)$ as $L \to \infty$, its leading coefficient could not be negative. An extension to a signed or complex observable must be named and derived separately. The historical negative-central-charge fits remain numerical records with their own assumptions.

10 Independent audit of the incoming disk extension

This section audits the preserved `matter-disk` sources at commit `881386a0ab00d917af6334a01423fb8b22469f5e`. Original scripts, outputs and claims remain unchanged in the incoming record. The findings below qualify their use in the present formal manuscript; they do not replace the source history.

10.1 Endpoint filtering changes the measured massive-band exit

The incoming second-model experiment uses

$$h(k) = (t_1 + \cos k)\sigma_x + \sin k\, \sigma_y + (m + i\gamma)\sigma_z$$

and compresses its lower-band biorthogonal projector to a block. Let ν_j be the compressed eigenvalues. For an endpoint filter $\delta > 0$, the retained set is

$$\mathcal{J}_\delta = \{j : |\nu_j| > \delta,\ |1 - \nu_j| > \delta\}, \qquad R_\delta = \max_{j \in \mathcal{J}_\delta} \frac{|\operatorname{Im} \operatorname{Log}((1-\nu_j)/\nu_j)|}{\pi}.$$

The logarithm here is principal. The reported exit is the crossing of $R_\delta = 1/2$ returned by a geometric scan and bisection, following the source algorithm. The scan is not a proof of a unique crossing or of the first crossing over a continuous parameter interval. The filter is part of the instrument.

An independent implementation kept the model, antiperiodic ring $N = 2000$, block length $L = 40$ unit cells and $t_1 = 0.5$ fixed. It constructed each band projector as $(I - h/E)/2$, with principal $E = \sqrt{d \cdot d}$, then assembled the block through Fourier coefficients. This is independent of the source's left/right eigenvector assembly. Both constructions agree at the checked band points to floating-point precision. The protocol was recorded before execution.

Endpoint filter δ	Massive $m = 0.4$: returned γ	Massive $\langle K \rangle$	Massless $m = 0$: returned γ	Massless $\langle K \rangle$
10^{-6}	0.271636929	1.057972846	0.474341649	2.000000002
10^{-8}	0.271636929	1.057972846	0.474341649	2.000000002
10^{-9}	0.169169932	1.022352491	0.474341649	2.000000002
10^{-10}	0.169169932	1.022352491	0.474341467	1.999995744
10^{-12}	0.122791366	1.011745220	0.474227956	1.997345192

The shipped 10^{-9} massive result is reproduced. Its stability under increasing block size did not establish stability under changing the endpoint filter. The large massive-band change between 10^{-8}, 10^{-9} and 10^{-12} prevents identifying that returned number with a filter-independent physical threshold on the evidence presently available.

The massless result is stable at the displayed precision for filters 10^{-6} through 10^{-9}. At smaller filters it also shifts. The very small endpoint eigenvalues belong to a nonnormal eigenproblem; neither that shift nor a small residual alone resolves its exact-arithmetic limit.

At the ten reported parameter values, the maximum normalized eigenpair residual was 4.41×10^{-15}, using unit eigenvectors and normalization $\max(1, \|C_A\|)$. Eigenvector condition numbers ranged from approximately 135 to 575. These diagnostics were added after observing the sensitivity and are recorded as post-analysis diagnostics, not as prospective decision tests. A small residual is not a validated eigenvalue-error bound for a nonnormal matrix.

What remains supported. The incoming universal statement that all tested opened bands exit at $\langle K \rangle = 2$ remains falsified at every massive filter in this table. The restricted massless coincidence has finite numerical support at specified filters. The exact Petermann identity and its unbroken-SSH band average are separate analytic results. Neither their derivation nor this table proves the equivalence between that average and the spatial-census exit.

What remains open. A filter-independent massive exit, if one is intended, needs a definition and a controlled limit involving δ, ring size, block size and numerical error. The order of limits must be stated. It should not be replaced by another fitted constant at a single filter. All failed universal-law claims and original controls remain part of the record.

Reproduction files are `review/INCOMING_EXIT_FILTER_PROTOCOL.md`, `review/incoming_exit_filter.py`, `review/incoming_exit_filter_results.json`, `review/incoming_exit_residuals.py` and `review/incoming_exit_residuals.json`.

10.2 The algebra that survives a biorthogonal projector

For a general idempotent block matrix, the identity $C_{AB}C_{BA} = C_A - C_A^2$ and its polynomial spectral mapping survive without Hermiticity. Therefore the branch-independent radial identity

$$\sqrt{|1 - 4\nu(1-\nu)|} = |1 - 2\nu|$$

legitimately transfers the complex radial disk count. This is proved in D3–D4. It is not the same as transferring positivity, singular-value counting or the Gaussian number-variance interpretation. The exact idempotent

$$C = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}, \qquad C^2 = C,$$

has $C_A = 2$ and $C_{AB}C_{BA} = -2$. Thus idempotence by itself permits a negative cross product and radial value 3. Substituting $C_{AB}C_{AB}^* = 1$ silently changes the observable and loses the identity. A positive metric, if introduced, must also specify the adjoint and orthogonal partition in that metric.

For a Hermitian positive contraction, the unsigned boundary coordinate $\Gamma_b = \sqrt{1 - 4\lambda}$ is valid and lies in $[0, 1]$. In a pure state it equals $|1-2\nu|$, not the signed coordinate itself. A zero threshold count means all strengths are below the chosen threshold; it does not mean the cross block vanishes. The warm uniform-ring row has zero count but positive trace, making this distinction observable within the existing calculation.

10.3 Complex poles, coordinate branches and physical interpretations

For $z \neq -1$, the Cayley coordinate obeys

$$\left|\frac{z-1}{z+1}\right| < 1 \quad \Longleftrightarrow \quad \operatorname{Re} z > 0.$$

Consequently, complex width alone does not imply an outside-disk point. The explicit family $z = 1 - i\gamma$, $\gamma > 0$, gives

$$\Gamma = \frac{-i\gamma}{2 - i\gamma}, \qquad |\Gamma|^2 = \frac{\gamma^2}{4 + \gamma^2} < 1.$$

An opened or resonant system needs a specified map to z and a condition on its real part before an outside-disk conclusion follows. The complex logarithm also requires a branch: for arbitrary

W, the condition is $\cos(\operatorname{Im} W) > 0$; the interval $|\operatorname{Im} W| < \pi/2$ is equivalent only for the principal branch.

Raw Hamiltonian-energy ratios have an additional reference issue. Energies $(1, 2)$ have ratio 2 and coordinate $1/3$. Shifting both by -1.25 gives $(-0.25, 0.75)$, ratio -3 and coordinate 2, while the energy gap remains one. This is a counterexample to an energy-zero-independent interpretation of arbitrary eigenenergy ratios. It is **not** a criticism of VOTKP's explicitly defined removal energies, whose physical references are part of their definitions.

A negative coefficient inferred from a biorthogonal Rényi-entropy fit likewise does not establish a negative coefficient for a literal nonnegative channel cardinality. If $N(\xi) = a \log \xi + O(1)$ with $a < 0$ as $\xi \to \infty$, then $N(\xi)$ eventually becomes negative, contradicting an unsigned count. A signed or otherwise redefined observable needs its own definition and proof. The positive-covariance entropy/count relationship cannot be continued into an indefinite or nonpositive setting by terminology alone. The actual supplied Rényi-fit outputs remain preserved as numerical results under their stated biorthogonal prescription.

10.4 The gas inverse survives with its required state information

The incoming helium difference identity

$$\log \frac{n_0 n_2}{n_1^2} = -\frac{I_1 - I_0}{k_B T} - \log 4$$

is population-only under the declared ground partitions. Its sum identity

$$\log \frac{n_0}{n_2 n_e^2 \lambda_e^6} = \frac{I_0 + I_1}{k_B T} - \log 4$$

is also correct, but its evaluation requires $n_e \lambda_e^3$ for each sample. It is density-compensated, not obtainable from charge fractions alone. In a pure neutral species, n_e may be inferred from fractions **and an absolute nucleus density**; fractions alone do not provide it. Temperature-dependent partitions add terms to both slopes. G8–G9 prove the general inverse, its partition correction and its deterministic error bounds.

Generating synthetic populations from a supplied atomic ladder and reconstructing that same ladder is an identity and implementation check. It is not independent experimental confirmation of the threshold prediction. This scope leaves the inverse construction intact and identifies what a real measurement must supply.

11 What was reproduced, and what changed

11.1 The preceding review remains part of the record

The complete preceding review archive is preserved as `archive/preceding_review_reproduction.zip`, with its own manifest and executable reproduction entrypoint. It contains the recovered Matter v5 and VOTKP v2.9 papers and source archives, the thermal-boundary controls, the helium bridge, and the published xenon source-data audit. The present edition extends that review; it does not discard its inputs or quietly replace its negative controls.

The earlier thermal and helium suites contain 22 passing checks: ten thermal-boundary controls and twelve helium/VOTKP controls. These are checks of specified implementations, not 22 experimental confirmations. The original Ising fit checks reused identified cached results; they did not rerun every lattice-size sweep. The independent Majorana coefficient convention still needs reconciliation before equating coefficients from different counting prescriptions. A sign correction in the prior source analysis is preserved rather than erased from the history.

Using the threshold inputs $I_0 = 24.587389011$ eV and $I_1 = 54.4177655282$ eV gives the anchored helium coordinate $\Gamma = 0.37757506698375054$ and binding sum 79.0051545392 eV. The input file retains its experimental/theoretical source labels. At nucleus density 10^{22} m^{-3}, the ground-partition ideal Saha model gives neutral fractions approximately 0.999374, 0.289781 and 8.73350×10^{-6} at 10,000, 20,000 and 50,000 K respectively. The atomic threshold coordinate stays fixed while the gas state changes. This is the specific sense in which an atomic record alone does not determine a gas population.

The xenon audit reproduced an identified row in the 2020 source-data tables: 114.323 ± 35.0164 attoseconds at 75.36 eV, with the source's uncertainty convention retained. The original archive contains 75 CSV files; eight relevant displayed rows were audited. This is a published-data reproduction. It does not recover raw channel phases or uniquely reconstruct a memory kernel. Neither that numerical agreement nor a colour analogy supplies the missing channel weights.

11.2 New finite checks

The protocols, executable code, exact rational witnesses and full JSON records accompany the edition. A protocol recorded before the associated implementation ran is identified as prospective; historical-source reruns and follow-up residual diagnostics are identified as retrospective. No external preregistration or independent laboratory experiment is claimed.

Check family	Coverage	Result and boundary
Finite moments	Orders 1 through 12 in exact rational arithmetic; complex response, zero and reflected-intensity controls	All stated identities hold. Equal low moments do not determine a whole response.
Gas and response	18 flags, including 100 rational ladders, 64 neutrality systems and 16 inverse ladders of lengths 2–8	All pass within declared tolerances. These remain ideal-model checks.
Boundary adversarial suite	128 complex-contraction pairs, 455 threshold-margin checks and 64 near-projector cases	All pass; discontinuity at a threshold and non-Gaussian counterexamples are retained.
Gaussian information	64 covariance cases, sharp lower family and divergent upper-ratio family	Checks support the analytic bounds; no universal upper constant without a spectral gap.
Incoming §7 and §8	Exact source scripts at the supplied commit	Five §7 PASS groups and three §8 PASS groups; figure 02 regenerated.
Exit-filter audit	Five endpoint filters, both massive and massless families	Massive exit changes materially with the filter. A universal exit rule remains unsupported.

Check family	Coverage	Result and boundary
Lean core	34 declarations; five false controls	All 34 compile; all five altered claims are rejected on mathematical goals.

The finite-moment illustration uses the stronger perturbation $(3, 7, 1, 5)/16$, whose third moment is 3. The separately compiled Lean witness uses $(7, 11, 5, 9)/32$, whose third moment is $3/2$. Both share moments $(1, 0, 5)$ with the uniform distribution on $(-3, -1, 1, 3)$; they are different amplitudes of the same construction. They are not reported as the identical numerical instance.

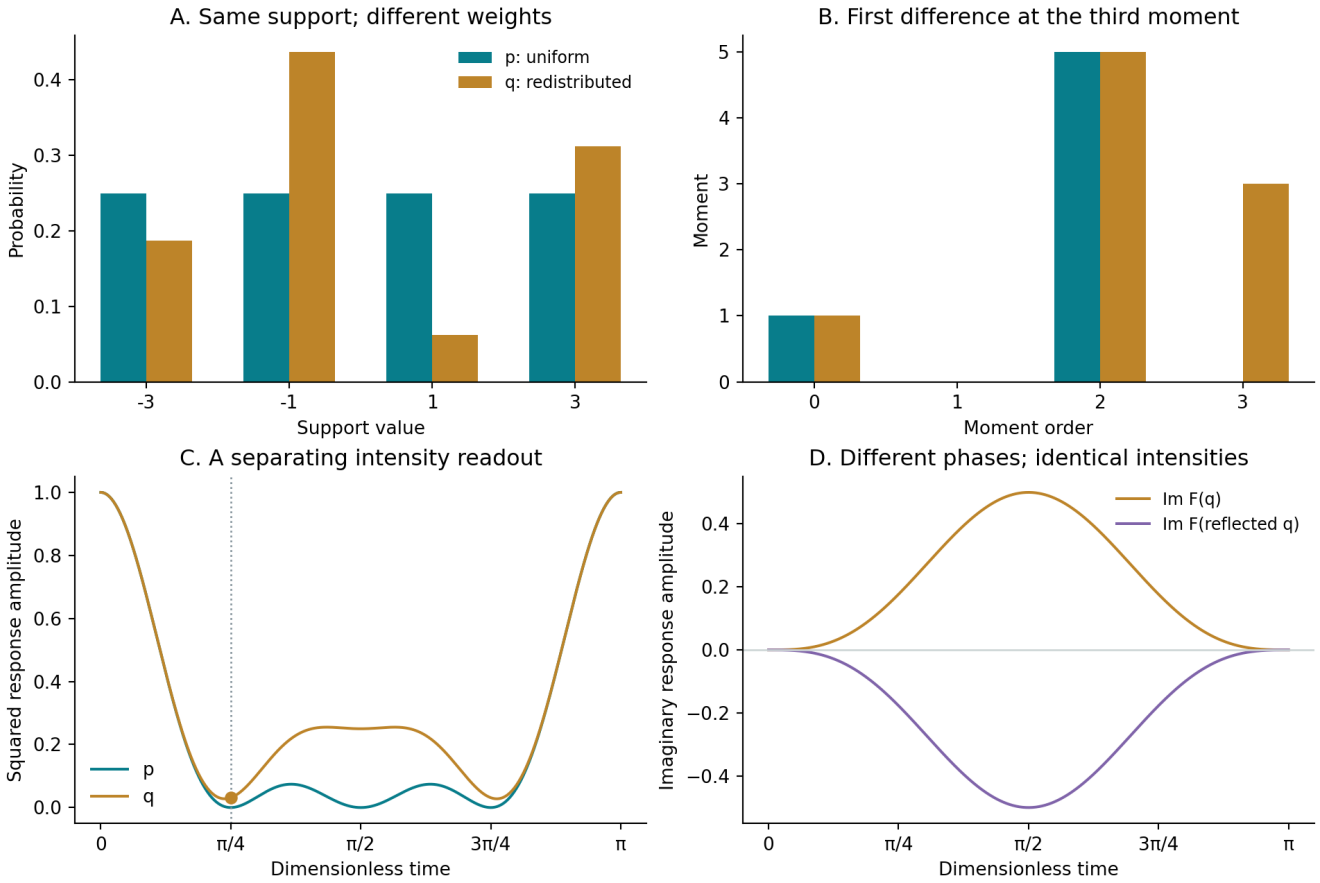


Figure 1: Equal low moments, different response. The reflected control keeps intensity unchanged while reversing the phase information. All curves are evaluations of the specified finite measures.

11.3 The incoming gas opposite test is now resolved within its model

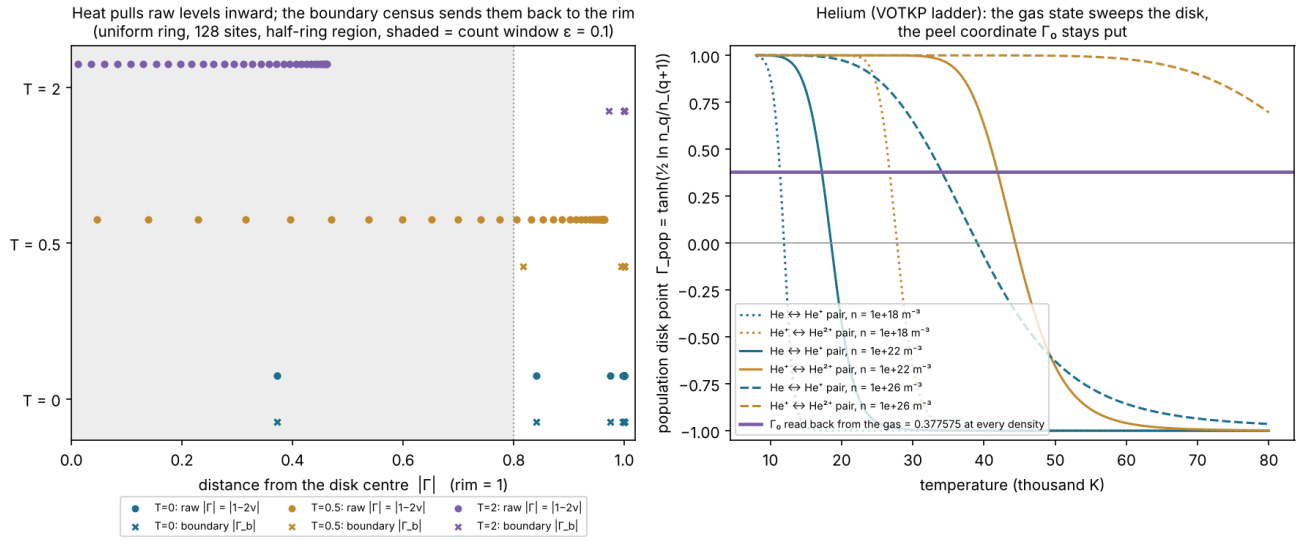


Figure 2: Preserved source figure 02 from the incoming commit. Left: raw local occupations and cross-boundary strengths use different disk coordinates at finite temperature. The warm zero count does not mean zero cross-boundary correlation. Right: the declared ideal gas populations vary while the input atomic peel coordinate stays fixed; reading it back is a synthetic consistency check. The figure's original wording is preserved, with these qualifications.

At 30,000 and 32,000 K, the ground-partition inverse returns $\Gamma = 0.37757506698375$ across the five tested densities 10^{18} through 10^{26} m^{-3}. Replacing the middle partition by the deliberately synthetic function

$$U_1(T) = 2 + 4\exp[-10\,\mathrm{eV}/(k_B T)]$$

while ignoring its temperature dependence returns $\Gamma = 0.36614943350748$. The bias is about -0.0114256334763. Subtracting the known partition terms restores the original coordinate within 1.433×10^{-14}. This is a direct negative control against an unjustified fixed-partition inversion. The synthetic excitation is not represented as a validated helium partition model.

The population triple ratio cancels density. The sum combination requires a measured or independently inferred $n_e\lambda_e^3$ for each state. This distinction survives the algebra and must survive any experimental proposal.

11.4 New conclusions with explicit limits

The Gaussian information bound has sharp lower constant 4; its upper bound needs occupations away from 0 and 1. The non-Gaussian control retains entanglement with a zero one-body cross block.

The gas inverse covers every finite positive ladder with partition corrections and conditioning bounds. Its required state inputs must be known; synthetic recovery is not an independent atomic prediction.

The finite-moment, equal-spectrum and free-streaming witnesses identify incomplete measured records. They do not claim that identical complete states evolve differently.

12 Relation to primary literature

The foundational tools used here have established histories. Reduced density operators from fermionic correlation matrices and free-system entanglement are developed by Peschel and by Peschel–Eisler [3,4]. Finite-temperature mutual information and area-law reasoning have a broader literature including Wolf, Verstraete, Hastings and Cirac [5]. The present finite proof for B13 is supplied directly, so its correctness does not depend on transferring a theorem from a different model class.

Chang, You, Wen and Ryu study entanglement and nonunitary criticality in non-Hermitian systems [6]. Their setting helps explain why a biorthogonal entropy need not behave like the entropy of a positive density operator. It does not make a negative coefficient into a negative unsigned census, or prove the exit rule considered here.

Two recent directions are relevant but remain distinct. Tong, Kuwahara and Gong's 2026 clustering theorem treats Bose–Hubbard Gibbs states and high-temperature correlation/mutual-information structure [7]. Those are interacting bosons with unbounded local occupation, whereas B1–B13 are finite covariance statements with a fermionic Gaussian interpretation. Kafle and colleagues' 2025 preprint reports spin and charge probes in a one-dimensional ^{6}Li gas [8]. Such separate readouts motivate specifying the response channel; they do not directly measure the present X_A or validate a VOTKP inversion.

Zhong and colleagues' xenon time-delay measurements [9] are the identified experimental source used in the preceding audit. Their phase-sensitive physical problem illustrates why threshold positions and channel response are different data. A calibrated response model and source-resolved amplitudes remain necessary before inferring a unique retained-memory kernel from delays.

The literature review supplies context and model boundaries, not an assertion that these standard constructions or every derived inequality are unprecedented. No priority claim is made for a result solely because it was derived in this session.

13 Open Math OS handoff

This edition is prepared for import, not submitted or accepted. Public pages for Open Math OS describe a collaborative research workspace with file/folder uploads, Python/LaTeX tools and Lean with Mathlib v4.34. No authenticated import was attempted, and the exact uploader limits, archive extraction behaviour and account quota remain untested. The package pins the version actually checked: Lean 4.34.1 and Mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`.

The reading files are the separate PDF and self-contained HTML. One reproduction ZIP contains the editable manuscript, proof sources, code, protocols, data, logs, source archives and integrity manifest. It does not wrap those two reading files into another combined download. `README_REPRODUCTION.md` is the entrypoint. Run `python reproduce_formal.py --require-manifest` to verify inputs, extract preserved sources, rerun the declared numerical suite and rebuild both reading formats. Add `--lean` for the pinned kernel check; initial dependency installation requires access to the official toolchain/cache. The historical source archives are complete, but not every historical computation is claimed rerun.

13.1 Bounded next research tasks

1. **Extend the formal spectral layer.** Prove B3, the ordered spectral comparison in B4 and the operator-norm stability in B7 using the current matrix core. Supply the exact min–max and perturbation hypotheses rather than assume the desired conclusion.
2. **Formalize the information bound.** Prove the trace Jensen step and B13 for finite matrices, then separately connect the trace functional to the declared Gaussian state. Preserve the non-Gaussian counterexample.
3. **Extend Saha existence formally.** Move from the compiled three-charge uniqueness result to general finite-Z existence and the variance derivative; keep positivity and the finite model assumptions explicit.
4. **Define the exit before fitting it.** Specify partition, coupling, endpoint treatment and order of limits. Use P3 and the filter audit as mandatory controls. A successful finite match at $\overline{K} = 2$ is not an acceptable theorem hypothesis.
5. **Test an empirical gas inverse.** Select a real partition dataset, quantify populations and electron-density uncertainties, and establish the equilibrium regime before fitting slopes. Predictions and rejection criteria should be fixed before examining held-out measurements.

These tasks do not authorize replacing the author's connection-energy framework with a convenient surrogate. Any extension claiming that connection must first establish its definitions and correspondence against the author's original sources.

14 References and source versions

1. J. Beasley, *Matter at a Scale*, v5. DOI: 10.5281/zenodo.22019893. Original paper and reproduction archive preserved.
2. J. Beasley, *VOTKP*, v2.9. DOI: 10.5281/zenodo.23340757. Original paper and code/proofs/data archive preserved. Incoming *matter-disk* source: commit 881386a, 3 November 2026.
3. I. Peschel, "Calculation of reduced density matrices from correlation functions," *Journal of Physics A* 36 (2003), L205–L208. DOI: 10.1088/0305-4470/36/14/101; arXiv:cond-mat/0212631.
4. I. Peschel and V. Eisler, "Reduced density matrices and entanglement entropy in free lattice models," *Journal of Physics A* 42 (2009), 504003. arXiv:0906.1663.
5. M. M. Wolf, F. Verstraete, M. B. Hastings and J. I. Cirac, "Area Laws in Quantum Systems: Mutual Information and Correlations," *Physical Review Letters* 100 (2008), 070502. DOI: 10.1103/PhysRevLett.100.070502.
6. P.-Y. Chang, J.-S. You, X. Wen and S. Ryu, "Entanglement spectrum and entropy in topological non-Hermitian systems and nonunitary conformal field theory," *Physical Review Research* 2 (2020), 033069. DOI: 10.1103/PhysRevResearch.2.033069; arXiv:1909.01346.
7. X.-H. Tong, T. Kuwahara and Z. Gong, "Clustering Theorem for Bose–Hubbard Class Gibbs States," *Communications in Mathematical Physics* 407 (2026), 245, published 28 October 2026. DOI: 10.1007/s00220-026-05730-1.
8. A. Kafle, R. Senaratne, D. Cavazos-Cavazos, H.-Y. Cui, T. Giamarchi, H. Pu, X.-W. Guan and R. G. Hulet, "Low Energy Excitations of a 1D Fermi Gas with Attractive Interactions," preprint submitted 9 December 2025. arXiv:2512.08866.
9. S. Zhong and colleagues, "Attosecond electron–spin dynamics in Xe 4d photoionization," *Nature Communications* 11 (2020), 5042. DOI: 10.1038/s41467-020-18847-1; full primary article and source-data links. Original source-data archive retained.
10. Open Math OS and About Open Math Model, public pages checked 3 November 2026. Full readiness notes and access limitations are preserved in `platform/`.

15 Machine-checked scope — Matter, Gases and VOTKP

Status: **34 positive theorem declarations compile under Lean 4.34.1 and pinned Mathlib; five intentionally false controls are rejected.** The final executable verification record is `logs/verification.json`. Exact theorem statements are in `theorem_manifest.json`, and Lean's own elaborated types plus each declaration's axiom dependencies are printed in `logs/axioms.log`.

This is a formalization of a defined mathematical core. It does not certify every statement of the manuscript, the physical applicability of a model, empirical data, scientific novelty, or priority. No original VOTKP historical Lean suite is claimed to have been rebuilt here.

15.1 Checked mathematical content

Source module	Exact coverage	Manuscript relationship
`MatterGas/Boundary.lean`	Complex finite-matrix block defect identity; positive defect for a positive contraction; positive restricted defect and cross block product; `0 ≤ BB* ≤ A−A² ≤ I/4`; pure-projector reduction; disconnected zero; trace zero iff cross block zero	Full algebra/positivity core of B1–B2, algebraic part of B4, zero criterion in B5. Singular-value identification, ordered spectral comparisons and Gaussian interpretation remain manuscript proofs.
`MatterGas/Scalar.lean`	Cayley ratio scale invariance, strict bounds, inverse and anchored reconstruction; scalar covariance factorization, threshold equivalence and quarter bound; finite nonnegative-list threshold-count bound; disconnected scalar witness	Scalar ingredients of G1, B4–B5. The list bound is not presented as a standalone formalization of the matrix spectral theorem.
`MatterGas/Saha.lean`	Positive three-charge mean denominator; strict decrease of mean charge with electron density; uniqueness among positive neutral solutions; `0 < meanCharge < 2`; two population-ratio cancellation identities	Three-charge specialization of the uniqueness argument, plus the exact rational algebra of the §8 gas-peeling inverse. General finite-Z existence, derivative/variance identities and LTE physics remain manuscript arguments.
`MatterGas/Witness.lean`	A finite-difference vector annihilates every quadratic polynomial; two explicit positive distributions on velocities `(-3,-1,1,3)` share moments 0, 1, 2 and differ in moment 3 by `3/2`	Exact finite witness used in M1–M2. It does not prove the general order-p finite-difference theorem or a transport dynamics model.
Additional declarations in `Boundary.lean`	`1−4ν(1−ν)=(1−2ν)²` over complex numbers; every square matrix A embeds into idempotent `[[A,I],[A−A²,I−A]]`	Algebraic core supporting D1/D3/D5. It does not supply branch-independent spectral counts, positivity of a non-Hermitian idempotent, or the full Cayley branch theorem.

B3, B6–B13, remaining spectral parts of D1–D7, the collision/dynamical results K1–K4, general-Z gas results, response laws R1–R4 and Petermann statements P1–P2 are **not claimed machine-checked by this package**. Their manuscript proof and test status must be reported separately.

15.2 Trust and negative controls

Positive sources contain no `sorry`, `admit`, custom `axiom`, or `native_decide`. The verifier scans those sources, builds the package, requests `#print axioms` for every manifest declaration, checks exact coverage and rejects dependencies outside `propext`, `Classical.choice`, and `Quot.sound`. These are standard Lean/Mathlib foundational axioms, not additional physical assumptions.

The five false-control files must fail with an **unsolved mathematical goal**. A missing import or toolchain error does not count as a passing control. The rejected alterations claim that a common scale changes the Cayley ratio, disconnected thermal variance is zero, the maximum covariance defect is at most 1/5, the equal first three moments determine the third, or mean charge increases with electron density. The last control is stated at explicit exact positive parameters.

The ordinary build intentionally excludes `FalseControls/`. Its files are only run separately by the verifier, where failure is required. Positive definitions and all 34 theorem statements remain inspectable, rather than hidden behind an opaque numerical-success flag.

15.3 Pinned environment and runtime note

- Lean: `leanprover/lean4:v4.34.1`, official commit `5045d0056413266e57c625dcd7c365b10e377c52`.
- Mathlib: tag `v4.34.1`, exact commit `d13f23b723b8a846827a245b89c10fc7d3f11612`.
- Transitive dependency revisions: `lake-manifest.json`.
- Host: Linux, official release binaries and official Mathlib cache.

This execution container permits `/proc/self/exe` but does not resolve `/proc/<the same process's numeric PID>/exe` for Lean's path lookup. The documented optional `tools/self_exe_compat.c` redirects only that exact own-process alias to `/proc/self/exe`. It does not change Lean's kernel, axioms, tactic code or theorem statements, and it does not redirect another process's path. The verification JSON records whether this compatibility shim was present. Standard environments should run without it.

The toolchain binaries, cache, temporary setup attempts and build artifacts are not part of the distributable source package. A fresh standard environment must obtain the pinned public dependencies once, then run the provided verifier.

16 Scoped dependency audit following the reported ledger-implementation dispute

Audit date: 3 November 2026. This is a local source-and-dependency inspection, not a new experiment, a new assessment of the intended framework, or a reconstruction of another session's timing or billing. No tests were run for this audit. Historical files remain unchanged.

16.1 Finding

The named disputed runs have not been identified in the searched local material. The current package does inherit the preceding review as a runnable archive, and that review contains standard-model calculations described in places as repairs or a replacement thesis. Those descriptions need an explicit framework-fidelity qualification. Mathematical correctness of a specified covariance, Saha, Hermite, or finite-moment model does not establish that it implements the author's Direction–Spin–Throughput and connection-energy construction. Nor does a failure or limitation of one of those restricted models reject that construction.

The supplied alert is treated as a user-reported dispute about another implementation. This audit neither confirms its reported times nor attributes the disputed runs to the current project. It also does not establish that the relevant original package was read by another assistant. No exact package-to-code mapping for that framework is present in the reviewed mathematical notes. Framework fidelity therefore remains **unverified**, rather than verified, falsified, or disproved by the present controls.

16.2 Search scope and name collisions

The first bounded search covered 222 text/code files in the current formal tree and 21 in the preceding review, excluding generated artifacts, duplicated relocated/package-test trees, runtime/toolchain directories, raw HTML, and the original source directory. The source follow-up covered 593 text/code/JSON files under the preceding `sources/` and 182 under current `incoming_disk/` (the latter overlaps the first search). Parsed string values in `incoming_disk/metadata/drive_text_payloads.json` were searched separately without printing their payloads. Searches used word boundaries and the labels `E9`, `Heegner`, `midrange`, `split-at-zero`, `rally`, `Direction–Spin`, `Throughput`, `04:25`, `05:45`; related `connection energy`, `two books`, `two ledger`, `single clock`, `population`, `commutator`, and repair language was inspected selectively. This is a local search statement, not a claim that the named work does not exist elsewhere.

Term or reported run	Local result	Interpretation
E9 repairs	A lowercase `e9` occurs at `sources/VOTKP_v2_9_Code_Proofs_Data_20261013/VOTKP_v2_9_Code_Proofs_Data_20261013/work/rhythm/blockB.py:99–100` in the preceding project.	It is the error for a nine-octave leave-one-out feature model (`T18`), paired with `e8`. Nothing in these lines identifies it with the reported E9 repair. Do not conflate them.
Heegner helper, midrange, split-at-zeros, rally	No exact label found in the scoped files.	No local run-to-claim dependency identified. Not an exoneration of an unavailable implementation.
Direction–Spin–Throughput package / connection-energy note	No document matching those titles was identified in these local trees. A connection-energy phrase occurs in a separate Penrose comparison note.	An incidental phrase is not the specified package or evidence that its full implementation was read.
Throughput	Appears in Matter v5 and preserved Atlas source interpretations; also as neutral-filter optical throughput.	These are different usages. They cannot all be classified as the disputed ledger run.

Term or reported run	Local result	Interpretation
Commutator	Used explicitly by the preceding eight-mode Hermite closure and by normality discussion.	This is a bounded standard linear model, not an identified rally/commutator implementation.
04:25–05:45	No exact markers found.	Timing and billing not independently verified.

Paths in the following tables are relative to their named project root. Line references describe the files at the time of this inspection and can shift after annotation.

16.3 Actual inheritance and executable dependencies

Location	What is imported or executed	What the dependency does and does not establish
Current `reproduce_formal.py:169–172`	Extracts `archive/preceding_review_reproduction.zip`; runs its `reproduce_all.py` unless `--skip-baseline`.	The older standard-model review is an actual execution dependency of the default runner. A successful run establishes reproduction of that baseline, not framework fidelity.
Current `reproduce_formal.py:173–182`	Extracts the pinned `votkp-lean` commit and checks selected incoming script output.	Named upstream calculations are preserved and rerun. Output checks are not a proof of all historical interpretation.
Current `reproduce_formal.py:186–194`	Runs new explicit scripts and reads their PASS/check fields.	Checks conditional finite-model claims. These fields must not be described as validation of the complete ledger programme.
Current `README_REPRODUCTION.md:21,54,68`	Describes the prior archive, its default rerun, and inherited original paper/code/data inputs.	The README already says synthetic examples and Saha reconstruction are not new measurements. Add the distinct qualification that reproduction does not establish the missing original-framework mapping.
Preceding `reproduce_all.py:28–30,178–180`	Executes `tests/thermal_census.py`, the Saha bridge, and the Matter audit; reads the thermal checks.	The current default runner therefore transitively executes these bounded models.
Preceding `build_review.py:20–21,125,130`	Reads thermal/audit JSON; prints results and normality interpretation.	These are the concrete routes from model output to reader-facing claims. Preserve the historical output, but qualify its interpretation as below.
Current `gas/verify_formal_gas.py:1–18,26–48`	Uses its own explicit Saha equations, NumPy/SciPy, and specified inputs.	No import of an identified E9/Heegner/rally run was found. This code is a Saha-model implementation; it is not evidence of a two-ledger implementation.
Current `tests/moment_controls.py`, `review/adversarial_checks.py`, `review/gaussian_information_checks.py`, `lean/MatterGas/*.lean`	Explicit finite constructions, matrix identities, and conditional Lean theorems.	No named disputed-run import found. Algebraic correctness and source-faithful physical interpretation are separate audit questions.

16.4 Places requiring interpretation changes

The strongest issue is not a demonstrated wrong matrix identity. It is possible overextension from a standard surrogate to the author's intended framework.

Exact location	Existing claim or framing	Required treatment
Preceding `reviews/matter_audit.md:14,107–120`	"useful repair," "Strongest repair preserving the work," "Replace," and "Suggested replacement thesis."	Preserve as historical proposed standard-model continuation. Withdraw any implication that this replaces or corrects the complete Direction–Spin–Throughput/connection-energy framework. No source-faithful mapping has been demonstrated here.
Preceding `reviews/matter_audit.md:32–34,130`	Gas-at-rest normality and entropy-production qualifications; an eight-mode Hermite/BGK example.	The operator-dependent mathematical cautions stand. Explicitly limit the counterexample to the declared operator and norm; it does not decide the author's intended ledger variables or connection-energy observable.
Preceding `build_review.py:128–131,166`	General-sounding conclusions adjacent to the Hermite result.	Add that the example answers only a literal unrestricted normality claim for standard kinetic generators. It is not a tested realization or rejection of the intended framework.
Preceding `reviews/votkp_gas_bridge.md:130–139`, especially `:138`	Historical failure list and "no scalar-only … clock … bound"; a zero commutator for a closed Hermitian atomic Hamiltonian.	Preserve as reports about the named restricted records/models. The zero commutator is correct for that Hamiltonian, but does not identify the user's connection energy with that commutator. No inference against a richer direction/spin/throughput record is justified without deriving the mapping.
Preceding `build_review.py:52,116`	Existing sources "contain … needed repair"; block identities "repair the proposed extension."	Qualify this as the proposed positive-covariance continuation under declared assumptions, not a repair of every part of the author's programme.
Current `manuscript/front.md:5,9,23,44`	Distinct objects, standard models, evidence classes, and chapter plan are already explicit.	Add the compact dependency notice below, so readers do not infer that conditional formalization certifies framework fidelity.
Current `review/formal_review.md`	Analytical approval of B/K/D/G/R/M/P statements.	Append that approval is of displayed hypotheses and proofs, not a certification that the user's intended ledger model has been implemented. This addendum can be referenced rather than rewriting every theorem.

The relevant original Matter source itself identifies the dynamic paragraph as an analogy: preceding `sources/Matter_v5_release/matter_at_a_scale_v5.tex:538–540`. The ensuing normality

prose is at :540–555, with throughput and entropy production at :575–581. That matters: the review can flag missing hypotheses in literal broad claims without pretending that a surrogate calculation falsifies the intended construction. The original hypothesis/analogy marking must travel with any quotation or summary.

16.5 Current bounded models that can remain, with their proper status

- Current `math/boundary_theorems.md:3` already identifies standard finite-dimensional mathematics and distinguishes analytic proof from Lean verification. Its positive-covariance statements remain conditional theorems; they neither encode nor refute all ledger variables.
- Current `gas/formal_gas_response.md:3,47,187,207,349` explicitly bounds the finite Saha model, its temperature response, scale covariances, and implementation checks. `gas/gas_peel_inverse.md:49,77` correctly distinguishes a synthetic inverse identity and an excited-partition control from measurement. These can remain as **standard equilibrium benchmarks**, without claiming that a population vector is the whole intended state.
- Current `math/moment_matched_transport.md:176` says exactly what its construction establishes: density–flow–stress alone fails to determine later force-free transport for arbitrary positive velocity distributions. Its separate erasure toy is not a reconstruction of the user's dynamics. The result concerns omitted higher moments, not an argument that direction/spin/throughput is ineffective.
- Current `manuscript/moment_records.md:31,41` restricts non-determination to the stated finite moment record and labels the finite-difference construction standard. These statements can remain; a richer record is outside their no-go hypothesis.
- Preceding `tests/thermal_census.py:42,69–83` states its synthetic scope and implements the explicit matrix `A=-Q-iM`, with `Q=diag(0,0,0,1,1,1,1,1)`. The calculation establishes nonnormal contraction in that fixed weighted finite model. Neither population bookkeeping nor its one collision-time scale establishes fidelity to a multiple-book construction.

The notice recommended by this audit is applied in the opening source-integrity section. The full audit, including its suggested wording, remains in the reproduction archive.

16.6 Decision

Keep the exact proofs, declared finite-model outputs, original inputs, and failed historical hypotheses. Mark the framework attribution as unresolved and the affected replacement language as historical/disputed. Do not publish the questioned interpretation as settled, do not silently delete provenance, and do not run another surrogate experiment as a substitute for reading and mapping the intended model. This audit makes no finding about another session's timing, billing, or culpability.

17 Appendix: exact Lean declaration inventory

These are the exact 34 source statements, with shared declarations and definitions in their named modules. Their source proofs and elaborated types are in the ZIP. The printed axiom audit is the compilation record. P3 and the full analytic manuscript are not claimed machine-checked.

L1. block defect identity

Module: `MatterGas/Boundary.lean`.

```
theorem block_defect_identity (C : Matrix (m ⊕ n) (m ⊕ n) ℂ) (hC : C.IsHermitian) :
    localBlock C - localBlock C * localBlock C = thermalDefect C + boundary C
```

L2. boundary posSemidef

Module: `MatterGas/Boundary.lean`.

```
theorem boundary_posSemidef (C : Matrix (m ⊕ n) (m ⊕ n) ℂ) :
    (boundary C).PosSemidef
```

L3. contraction defect posSemidef

Module: `MatterGas/Boundary.lean`.

```
theorem contraction_defect_posSemidef (C : Matrix (m ⊕ n) (m ⊕ n) ℂ)
    (h0 : 0 ≤ C) (h1 : C ≤ 1) : (C-C*C).PosSemidef
```

L4. thermalDefect posSemidef

Module: `MatterGas/Boundary.lean`.

```
theorem thermalDefect_posSemidef (C : Matrix (m ⊕ n) (m ⊕ n) ℂ)
    (h0 : 0 ≤ C) (h1 : C ≤ 1) : (thermalDefect C).PosSemidef
```

L5. boundary le local defect

Module: `MatterGas/Boundary.lean`.

```
theorem boundary_le_local_defect (C : Matrix (m ⊕ n) (m ⊕ n) ℂ)
    (h0 : 0 ≤ C) (h1 : C ≤ 1) :
    boundary C ≤ localBlock C - localBlock C * localBlock C
```

L6. thermalDefect projector

Module: `MatterGas/Boundary.lean`.

```
theorem thermalDefect_projector (C : Matrix (m ⊕ n) (m ⊕ n) ℂ) (hp : C*C=C) :
    thermalDefect C = 0
```

L7. boundary projector

Module: `MatterGas/Boundary.lean`.

```
theorem boundary_projector (C : Matrix (m ⊕ n) (m ⊕ n) ℂ)
    (hC : C.IsHermitian) (hp : C*C=C) :
    boundary C = localBlock C - localBlock C * localBlock C
```

L8. boundary disconnected

Module: MatterGas/Boundary.lean.

```
theorem boundary_disconnected (A : Matrix m m ℂ) (D : Matrix n n ℂ) :
    boundary (Matrix.fromBlocks A 0 0 D) = 0
```

L9. boundary trace zero iff

Module: MatterGas/Boundary.lean.

```
theorem boundary_trace_zero_iff (C : Matrix (m ⊕ n) (m ⊕ n) ℂ) :
    (boundary C).trace = 0 ↔ crossBlock C = 0
```

L10. local defect le quarter

Module: MatterGas/Boundary.lean.

```
theorem local_defect_le_quarter (A : Matrix m m ℂ) (hA : A.IsHermitian) :
    A-A*A ≤ (1/4:ℝ) • (1 : Matrix m m ℂ)
```

L11. boundary bounds

Module: MatterGas/Boundary.lean.

```
theorem boundary_bounds (C : Matrix (m ⊕ n) (m ⊕ n) ℂ)
    (h0 : 0 ≤ C) (h1 : C ≤ 1) :
    0 ≤ boundary C ∧ boundary C ≤ (1/4:ℝ) • (1 : Matrix m m ℂ)
```

L12. square offset identity

Module: MatterGas/Boundary.lean.

```
theorem square_offset_identity (nu : ℂ) :
    1-4*(nu*(1-nu)) = (1-2*nu)^2
```

L13. arbitrary block idempotent

Module: MatterGas/Boundary.lean.

```
theorem arbitrary_block_idempotent (A : Matrix m m ℂ) :
    let C := Matrix.fromBlocks A (1 : Matrix m m ℂ) (A-A*A) (1-A)
    C*C=C
```

L14. meanCharge den pos

Module: MatterGas/Saha.lean.

```
theorem meanCharge_den_pos (a b x : ℝ) (ha : 0<a) (hb : 0<b) (hx : 0<x) :
    0 < x^2+a*x+b
```

L15. meanCharge strictAnti

Module: MatterGas/Saha.lean.

```
theorem meanCharge_strictAnti (a b x y : ℝ) (ha : 0<a) (hb : 0<b)
    (hx : 0<x) (hxy : x<y) : meanCharge a b y < meanCharge a b x
```

L16. saha neutrality unique

Module: MatterGas/Saha.lean.

```
theorem saha_neutrality_unique (a b N x y : ℝ) (ha : 0<a) (hb : 0<b) (hN : 0<N)
    (hx : 0<x) (hy : 0<y) (hxeq : x = N*meanCharge a b x)
    (hyeq : y = N*meanCharge a b y) : x=y
```

L17. meanCharge bounds

Module: MatterGas/Saha.lean.

```
theorem meanCharge_bounds (a b x : ℝ) (ha : 0<a) (hb : 0<b) (hx : 0<x) :
    0 < meanCharge a b x ∧ meanCharge a b x < 2
```

L18. saha ratio cancellation

Module: MatterGas/Saha.lean.

```
theorem saha_ratio_cancellation (n0 n1 n2 x S0 S1 : ℝ)
    (hn0 : n0 ≠ 0) (hn1 : n1 ≠ 0) (hx : x ≠ 0) (hS0 : S0 ≠ 0)
    (h01 : n1/n0 = S0/x) (h12 : n2/n1 = S1/x) :
    n0*n2/n1^2 = S1/S0
```

L19. saha product cancellation

Module: MatterGas/Saha.lean.

```
theorem saha_product_cancellation (n0 n1 n2 x S0 S1 : ℝ)
    (hn0 : n0 ≠ 0) (hn1 : n1 ≠ 0) (hx : x ≠ 0) (hS0 : S0 ≠ 0) (hS1 : S1 ≠ 0)
    (h01 : n1/n0 = S0/x) (h12 : n2/n1 = S1/x) :
    n0/(n2*x^2) = 1/(S0*S1)
```

L20. gamma scale

Module: MatterGas/Scalar.lean.

```
theorem gamma_scale (a b c : ℝ) (hc : c ≠ 0) :
    gamma (c*a) (c*b) = gamma a b
```

L21. gamma bounds

Module: MatterGas/Scalar.lean.

```
theorem gamma_bounds (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
    -1 < gamma a b ∧ gamma a b < 1
```

L22. gamma inverse

Module: MatterGas/Scalar.lean.

```
theorem gamma_inverse (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
    (1 + gamma a b) / (1 - gamma a b) = b / a
```

L23. gamma anchor reconstruction

Module: MatterGas/Scalar.lean.

```
theorem gamma_anchor_reconstruction (a b : ℝ) (ha : 0 < a) (hb : 0 < b) :
    a * ((1 + gamma a b) / (1 - gamma a b)) = b
```

L24. census factorization

Module: MatterGas/Scalar.lean.

```
theorem census_factorization (nu eps : ℝ) :
    nu*(1-nu) - eps*(1-eps) = (nu-eps)*(1-eps-nu)
```

L25. census threshold iff

Module: MatterGas/Scalar.lean.

```
theorem census_threshold_iff (nu eps : ℝ) (he : eps < (1:ℝ)/2) :
    eps*(1-eps) ≤ nu*(1-nu) ↔ eps ≤ nu ∧ nu ≤ 1-eps
```

L26. scalar defect bounds

Module: MatterGas/Scalar.lean.

```
theorem scalar_defect_bounds (nu : ℝ) (hlo : 0 ≤ nu) (hhi : nu ≤ 1) :
    0 ≤ nu*(1-nu) ∧ nu*(1-nu) ≤ (1:ℝ)/4
```

L27. threshold count trace bound

Module: MatterGas/Scalar.lean.

```
theorem threshold_count_trace_bound {ι : Type*} [Fintype ι]
    (lambda : ι → ℝ) (h : ∀ i, 0 ≤ lambda i) (theta : ℝ) (_ht : 0 < theta) :
    theta * ((Finset.univ.filter (fun i => theta ≤ lambda i)).card : ℝ)
      ≤ ∑ i, lambda i
```

L28. disconnected occupation witness

Module: MatterGas/Scalar.lean.

```
theorem disconnected_occupation_witness :
    (0:ℝ) < (3/10:ℝ)*(1-3/10) ∧ (3/10:ℝ)*(1-3/10) = 21/100
```

L29. quadratic moment null

Module: MatterGas/Witness.lean.

```
theorem quadratic_moment_null (a b c : ℝ) :
    -(a*(-3)^2+b*(-3)+c) + 3*(a*(-1)^2+b*(-1)+c)
      - 3*(a*1^2+b*1+c) + (a*3^2+b*3+c) = 0
```

L30. four moment weights

Module: MatterGas/Witness.lean.

```
theorem four_moment_weights :
    (1/4:ℝ)+(1/4)+(1/4)+(1/4)=1 ∧
    (7/32:ℝ)+(11/32)+(5/32)+(9/32)=1 ∧
    (0:ℝ)<7/32 ∧ (0:ℝ)<11/32 ∧ (0:ℝ)<5/32 ∧ (0:ℝ)<9/32
```

L31. four moment zero

Module: MatterGas/Witness.lean.

```
theorem four_moment_zero :
    fourMoment (1/4) (1/4) (1/4) (1/4) 0 =
    fourMoment (7/32) (11/32) (5/32) (9/32) 0
```

L32. four moment one

Module: MatterGas/Witness.lean.

```
theorem four_moment_one :
    fourMoment (1/4) (1/4) (1/4) (1/4) 1 =
    fourMoment (7/32) (11/32) (5/32) (9/32) 1
```

L33. four moment two

Module: MatterGas/Witness.lean.

```
theorem four_moment_two :
    fourMoment (1/4) (1/4) (1/4) (1/4) 2 = 5 ∧
    fourMoment (7/32) (11/32) (5/32) (9/32) 2 = 5
```

L34. four moment three separation

Module: MatterGas/Witness.lean.

```
theorem four_moment_three_separation :
    fourMoment (7/32) (11/32) (5/32) (9/32) 3 -
    fourMoment (1/4) (1/4) (1/4) (1/4) 3 = (3/2:ℝ)
```